Four-Force Generator Decomposition for Closed-Loop Systems: A Bottom-Up Protocol to Compile Ξ=(ρ,γ,τ) and Verify Control via Pump–Probe–Switch–Couple

0. Scope, claim level, and why this is *not* an ontology grab

This paper does **not** propose a new "what reality really is" story. It proposes a **portable operational routine** for comparing and controlling loop-bearing open systems using **intervention → response signatures** under a declared boundary and protocol.

0.1 Claim level: operational equivalence only

We define equivalence strictly in terms of **what you can do to a system** and **what reliably happens next**, not in terms of underlying ontology.

Let a system under a declared protocol P have a controlled generator family $\{\mathscr{L}_u\}$. For any loop L valid under P, we compile:

(0.1) $\Xi(L) = (\rho,\gamma,\tau)$

and we estimate local gains (MEP):

(0.2) $\Delta\Xi \approx G\,\Delta u$

and couplings:

(0.3) $\kappa_{A\leftarrow B}$ = (windowed causal influence estimate)

Operational equivalence between two systems S and S′ (under comparable protocols P and P′) means their *intervention–response signatures* match within tolerance, e.g.,

(0.4) $\|\Xi_S(L) - \Xi_{S'}(L')\| \le \varepsilon_\Xi$
(0.5) $\|G_S(L) - G_{S'}(L')\| \le \varepsilon_G$
(0.6) $\|\kappa_S - \kappa_{S'}\| \le \varepsilon_\kappa$

for appropriately matched loops (basis loops, Section 7).

No statement is made that their micro-constituents, "true forces," or metaphysical categories are the same—only that **they behave equivalently under the same class of interventions**.

0.2 EFT-style posture: valid inside declared boundary/protocol/timebase

The method is intentionally **effective-theory-like**: everything is conditioned on a declared boundary and sampling protocol:

(0.7) $P = (B, \Delta, h, \text{admissible } u)$

All derived objects—loops, signatures, Ξ, gains, κ—are **P-dependent**.

- If you change B (what counts as "inside"), the loop set may change.
- If you change Δ or h (timebase or measurement), the return map and signatures may change.
- If you change admissible u (what knobs you can turn), the gain artifact changes.

This is not a weakness; it is a guardrail. When the regime changes, we do not patch the narrative —we trigger falsifiability gates and route failure to actions.

Formally, the paper commits to the rule:

(0.8) Outside declared $P \Rightarrow$ treat mismatch as a failure event, not as "new interpretation"

Meaning: if stability gates fail, you do not salvage by reinterpretation; you execute a routing action (Section 8).

0.3 "No narrative patching": failure routes are part of the method

When gates fail, the protocol outputs one of five actions:

(0.9) Route $\in$ { shrink boundary, change proxy, split loop, change timescale, elevate model class }

This is the core reason the framework is not an ontology grab: it has **hard rejection behavior** and **explicit remediation paths** rather than unfalsifiable expansions.

0.4 What the paper actually delivers (portable bottom-up routine)

The deliverable is a **repeatable pipeline** that produces portable artifacts (Loop Cards and Coupling Cards) and a coupling graph that can be compared across domains.

The canonical routine (one line)

(0.10) four-force tagging → loop existence → Ξ-compression → operator probes → coupling κ → basis selection → falsifiability gates

Expanded (engineering steps)

1. **Four-force tagging (generator-level archetypes):** estimate signatures that separate potential/gradient, circulation/current, switching/jumps, and confinement/constraints.
2. **Loop existence:** accept a loop only if recurrence + return-map stability + bounded leakage pass.
3. **Ξ-compression:** compile $\Xi=(\rho,\gamma,\tau)$ from loop observables with explicit proxy choices and stability gates.

4. **Operator probes (MEP):** apply one-channel perturbations; reject regime shifts via jump/KL gates; estimate local gains G.
5. **Coupling κ:** estimate windowed causal influence κ_{A←B} (intervention-first; observational fallback labeled).
6. **Basis selection:** build the coupling graph; choose basis loops (recurrent SCC reps and/or signature span).
7. **Falsifiability gates:** enforce proxy drift, boundary accounting, probe backreaction, and control effectiveness; route failures by (0.9).

0.5 Safe reading guide for external audiences

If you are a physicist: treat this as a protocol for building **effective models** and **control summaries** of open systems with loop closure, not as a statement about fundamental interactions.

If you are an engineer: treat this as a way to produce **portable gain/coupling artifacts** with explicit rejection criteria—so you can compare systems and interventions without relying on domain-specific narratives.

If you are a theorist: the only "universal" claim is that **loop-bearing open systems** can often be meaningfully compressed and tested within a declared protocol. When they cannot, the method is designed to say so.

1. Starting point: a minimal dynamical object that physicists accept

This paper begins with the most conservative move: **treat an "object on the table" as an open system** obtained by drawing a boundary around some degrees of freedom, while everything outside the boundary enters only through **noise, drift, and discrete events**.

1.1 The object on the table = an open stochastic system on a state space $\mathcal{M}$

State space and stochastic structure

Let the system's microstate be x(t) living on a state space $\mathcal{M}$ (typically $\mathcal{M} \subseteq \mathbb{R}^d$ or a smooth manifold).

We assume a standard filtered probability space $(\Omega, \mathbb{F}, (\mathbb{F}_t)_{t\geq 0}, \mathbb{P})$ supporting:

- an m-dimensional Brownian motion W(t),
- a Poisson random measure N(dt,dy) on an event space $\mathcal{Y}$ with intensity ν(dy) dt.

Jump–diffusion model (physicist-friendly "minimal" form)

We model the open system as a jump–diffusion:

(1.1) $dx(t) = b(x(t),t)\,dt + \sigma(x(t),t)\,dW(t) + \int_{\mathcal{Y}} \kappa(x(t-),y,t)\,\tilde{N}(dt,dy)$

where:

- b(x,t) is the drift (systematic flow),
- σ(x,t) is the diffusion matrix (continuous fluctuations),
- κ(x−,y,t) is the jump size map (event impulse),
- x(t−) is the left-limit state just before a jump,
- $\tilde{N}$(dt,dy) is the compensated Poisson measure ($\tilde{N} = N - \nu(dy)dt$) so the jump term is "mean-zero" in the usual sense.

This representation is a deliberately broad "physics-accepting" umbrella:

- pure deterministic dynamics: set σ=0 and no jumps,
- Langevin/Fokker–Planck: no jumps,
- master equation / event-driven dynamics: diffusion small, jumps dominate.

Generator (so we can talk about "components" without metaphors)

For a smooth test function $\varphi: \mathcal{M} \to \mathbb{R}$, the (time-dependent) generator $\mathscr{L}_t$ is:

(1.2) $(\mathscr{L}_t \varphi)(x) = b(x,t)\cdot\nabla\varphi(x) + 1/2\ \mathrm{Tr}(\ a(x,t)\ \nabla^2\varphi(x)\ ) + \int_{\mathcal{Y}} [\varphi(x+\kappa(x,y,t)) - \varphi(x)]\ \nu(dy)$

(1.3) $a(x,t) = \sigma(x,t)\sigma(x,t)^{\mathrm{T}}$

Why the generator matters (marketing, but precise): **Protocol 2 will classify "force archetypes" at the level of $\mathscr{L}$**, because $\mathscr{L}$ is what you can *estimate* from trajectories + interventions, even when the microscopic ontology is unknown.

1.2 Boundary and protocol: what exactly is being held fixed?

A key move is to make "objecthood" operational by fixing a protocol.

Definition 1.1 (Protocol)

A protocol P is the tuple:

(1.4) $P = (B, \Delta, h, u)$

with:

- **B**: boundary specification (what is inside vs outside; what is treated as exogenous),
- **Δ**: sampling clock / observation cadence (continuous time t or discrete nΔ),
- **h**: observation map $h: \mathcal{M} \to \mathcal{Z}$ producing an observable macrostate z,
- **u**: admissible intervention channels (the knobs we are allowed to turn later).

Under P, the measured macrostate is:

(1.5) $z(t) = h(x(t))$ (continuous-time sampling)
(1.6) $z[n] = h(x(n\Delta))$ (discrete sampling, $n=0,1,2,\ldots$)

Important: In Protocol 2, a "loop" is never defined on x alone; it is defined on **(B,Δ,h)**. Change the boundary or sampling, and the loop classification may legitimately change.

1.3 What counts as a *loop*?

A loop is a **recurrent closure** observed under a fixed protocol P. Because the system is stochastic, we define loops in a way that remains meaningful under noise and jumps.

We give three equivalent (and practically usable) loop definitions.

Definition 1.2 (Loop as recurrent set in observable space)

Fix a metric d on the observable space $\mathcal{Z}$. Let $R \subseteq \mathcal{Z}$ be a region ("tube" or "cell"). Define the return time:

(1.7) $\tau_R = \inf\{ n \geq 1 : z[n] \in R \}$

We call R a **loop region** if:

(1.8) $\mathbb{P}(\tau_R < \infty \mid z[0] \in R) = 1$ and $\mathbb{E}[\tau_R \mid z[0] \in R] < \infty$

Interpretation:

- you always come back (recurrence),
- you come back on a finite typical timescale (not a one-off coincidence).

This is the most conservative definition: it is compatible with stationary stochastic dynamics and does not assume periodicity.

Definition 1.3 (Loop as approximate periodic closure with tolerance)

Sometimes we want the loop to look like "a cycle" to the eye. We say there is an ε-loop with period T (in samples) if:

(1.9) $d(z[n+T], z[n]) \leq \varepsilon$ for most n in a long window

A robust operational score for a candidate period T is the mean closure error:

(1.10) $M(T) = (1/(N-T)) \Sigma_{n=0}^{N-T-1} d(z[n+T], z[n])$

We say a **loop exists** if there is a T^* such that:

(1.11) $\min_{T \in [T_{min},T_{max}]} M(T) \leq \varepsilon$ and the minimizing T^* is stable across subwindows

This is "engineering-usable": you can compute it directly from logged data, even when the system has noise and occasional jumps.

Definition 1.4 (Loop as a stable return map)

Let s[n] be a scalar loop-phase coordinate extracted from z[n] (e.g., via a Poincaré section, a phase estimate, or simply a chosen event marker). A loop exists if the induced return map is stable:

(1.12) $s[n+T] \approx F(s[n])$ with F approximately time-invariant over the observation window

This definition is useful because Protocol 2 will later talk about "force archetypes" via how interventions distort the return map (continuous deformation vs discrete switching vs lock-in, etc.).

Minimal "loop acceptance" checklist (for later sections)

A loop candidate L under protocol P should satisfy all three practical conditions:

- **C1 (Closure):** $M(T^*)$ is small for some T^* (Eq. 1.10–1.11).
- **C2 (Recurrence):** returns occur repeatedly with finite typical return time (Eq. 1.8).
- **C3 (Protocol robustness):** small changes in Δ or ε do not destroy the loop (otherwise you discovered an artifact of sampling).

Once these conditions hold, we will treat the loop as a valid operational object for the rest of Protocol 2.

---

2. Four-force archetypes as a generator-level decomposition

We now introduce the **four-force archetypes** (G/E/W/S) as **operational components of the generator** $\mathscr{L}$.
This is not a claim about fundamental microphysics; it is a **computable split** of observed dynamics into four intervention-relevant "mechanism classes."

2.1 The generator $\mathscr{L}$ (recap, in the form we will decompose)

Assume x(t) follows the jump–diffusion model from Section 1. The generator acting on a smooth test function $\varphi: \mathcal{M} \to \mathbb{R}$ is:

(2.1) $(\mathscr{L}\varphi)(x) = (\mathscr{L}_{\text{drift}}\,\varphi)(x) + (\mathscr{L}_{\text{diff}}\,\varphi)(x) + (\mathscr{L}_{\text{jump}}\,\varphi)(x)$

(2.2) $(\mathscr{L}_{\text{drift}}\,\varphi)(x) = b(x)\cdot\nabla\varphi(x)$

(2.3) $(\mathscr{L}_{\text{diff}}\,\varphi)(x) = 1/2\ \mathrm{Tr}(\, a(x)\,\nabla^2\varphi(x)\,)$, $a(x)=\sigma(x)\sigma(x)^{\mathsf{T}}$

(2.4) $(\mathscr{L}_{\text{jump}}\,\varphi)(x) = \int_{\mathcal{Y}} [\varphi(x+\kappa(x,y)) - \varphi(x)]\,\nu(x,dy)$

Protocol 2 will split $\mathscr{L}$ (not the ontology) into four archetypes:

(2.5) $\mathscr{L} = \mathscr{L}_G + \mathscr{L}_E + \mathscr{L}_W + \mathscr{L}_S + \mathscr{L}_{\text{res}}$

where $\mathscr{L}_{res}$ is explicitly tracked residual (model error / unassigned structure), and the split is defined by **computable projections** and **estimable signatures**.

2.2 The computable split: one clean "default implementation"

A practical split that is both **physics-legible** and **data-computable** is:

- **S-like (Strong / confinement)**: fast restoring flow toward a constraint manifold (closure, lock-in).
- **G-like (Gravity / potential)**: gradient-like drift that admits a scalar potential (basin geometry).
- **E-like (EM / circulation)**: divergence-free probability-current component (loop circulation).
- **W-like (Weak / switching)**: jump kernel (triggered regime changes).

We implement this in two stages.

Stage A: isolate S-like confinement (constraints / lock-in)

Assume the system effectively lives near a constraint set (manifold) $\mathbb{N}$ defined by:

(2.6) $\mathbb{N} = \{ x \in \mathscr{M} : g(x)=0 \}$, $g:\mathscr{M}\to\mathbb{R}^k$

Let $J_g(x)$ be the Jacobian of g:

(2.7) $J_g(x) = \partial g/\partial x$

Define the normal projector (onto constraint-violating directions), assuming $J_g J_g^T$ is invertible on $\mathbb{N}$:

(2.8) $P_\perp(x) = J_g(x)^T (J_g(x) J_g(x)^T)^{-1} J_g(x)$

and the tangent projector:

(2.9) $P_\parallel(x) = I - P_\perp(x)$

Then define the **S-like restoring drift** as the normal "closure" component:

(2.10) $b_S(x) = P_\perp(x)\, b(x)$

and the remaining "within-loop" drift:

(2.11) $b_\parallel(x) = P_\parallel(x)\, b(x)$

Corresponding generator piece:

(2.12) $(\mathscr{L}_S \varphi)(x) = b_S(x)\cdot\nabla\varphi(x)$

Why this is computable: g(x) can be specified by design (hard constraints) or inferred (manifold learning / constraint regression), and $P_\perp$ is then explicit.

Stage B: split the within-loop drift into G-like vs E-like using probability current

Assume (locally or globally) that under the fixed protocol there exists a steady density $\pi(x)$ for the diffusion part (or an empirical stationary density). Define the probability current:

(2.13) $j(x) = b_{\|}(x)\,\pi(x) - 1/2\,\nabla\cdot(\,a(x)\,\pi(x)\,)$

(Here $\nabla\cdot(a\pi)$ is the vector with i-th component $\Sigma_j\,\partial(a_{ij}(x)\,\pi(x))/\partial x_j$.)

Then define:

(2.14) $b_E(x) = j(x) / \pi(x)$

(2.15) $b_G(x) = b_{\|}(x) - b_E(x)$

and the corresponding generator pieces:

(2.16) $(\mathscr{L}_E\,\varphi)(x) = b_E(x)\cdot\nabla\varphi(x)$

(2.17) $(\mathscr{L}_G\,\varphi)(x) = b_G(x)\cdot\nabla\varphi(x)$

Interpretation (tight, non-metaphorical):

- **E-like** is exactly "the part of drift that carries nonzero probability current around loops."
- **G-like** is "the remaining drift after removing that current," i.e., the part that can be represented by an effective potential under the chosen metric/weights.

Stage C: define W-like switching as the jump operator

From the jump kernel $\nu(x,dy)$, define the state-dependent jump intensity (hazard rate):

(2.18) $\lambda(x) = \int_{\mathcal{Y}}\,\nu(x,dy)$

and the normalized jump destination kernel:

(2.19) $K(x,dy) = \nu(x,dy) / \lambda(x)$ (when $\lambda(x)>0$)

Then define the **W-like generator**:

(2.20) $(\mathscr{L}_W\,\varphi)(x) = \lambda(x)\,\int_{\mathcal{Y}}\,[\varphi(x+\kappa(x,y)) - \varphi(x)]\,K(x,dy)$

This is "weak-like" in the operational sense: small continuous drift change is not the driver; **rare switching events** dominate macro reconfiguration.

2.3 How to estimate the split from logs / trajectories (minimal estimators)

Assume discrete sampling at step Δt with observed microstate $x[n]=x(n\Delta t)$. For a local neighborhood around x, estimate conditional moments:

(2.21) $\hat{b}(x) \approx (1/\Delta t)\,\mathbb{E}[\,x[n+1] - x[n] \mid x[n]\approx x\,]$

(2.22) $\hat{a}(x) \approx (1/\Delta t)\,\mathrm{Cov}[\,x[n+1] - x[n] \mid x[n]\approx x\,]$

Detect jump events by a discontinuity rule (one simple option):

(2.23) jump at $n \Leftrightarrow \|x[n+1] - x[n]\| > \theta(x[n])$

and estimate jump intensity:

(2.24) $\hat{\lambda}(x) \approx (1/\Delta t)\, \mathbb{P}(\text{ jump} \mid x[n] \approx x)$

Estimate empirical stationary density $\hat{\pi}(x)$ by histogram/KDE in low dimension, or by density-in-embedding for high dimension (the protocol decides the representation). Then estimate current:

(2.25) $\hat{j}(x) = \hat{b}_{\|}(x)\, \hat{\pi}(x) - 1/2\, \nabla\cdot(\, \hat{a}(x)\, \hat{\pi}(x)\,)$

and:

(2.26) $\hat{b}_E(x) = \hat{j}(x) / \hat{\pi}(x)$, $\hat{b}_G(x) = \hat{b}_{\|}(x) - \hat{b}_E(x)$

Constraint manifold estimation (if not given) can be operational: fit $\hat{g}(x)$ such that constraint residual is small and normal restoring is strong (see S-signature below), then use $\hat{P}_{\perp}$ built from $J_{\{\hat{g}\}}$.

2.4 Signatures: what each component "looks like" in data (estimable)

A **signature** is a statistic, geometric quantity, or topological invariant computed from logs/trajectories that (i) is stable under small protocol noise, and (ii) changes predictably under targeted interventions later (Pump/Probe/Switch/Couple).

Below are **minimal signature sets** that are both interpretable and computable.

G-like signatures (potential / basin geometry)

G1. Alignment with a gradient field
Compute a fitted potential $\hat{\Phi}$ (e.g., by solving a weighted Poisson/least-squares problem so that $-\nabla\hat{\Phi}$ best matches $\hat{b}_G$). A direct alignment score is:

(2.27) $A_G = \mathbb{E}[\, (\, \hat{b}_G\cdot(-\nabla\hat{\Phi})\,) \,/\, (\|\hat{b}_G\|\|\nabla\hat{\Phi}\| + \varepsilon_0\,)\,]$

A_G near 1 indicates strongly potential-like flow.

G2. Basin depth proxy (escape cost)
Choose an attractor neighborhood U and a boundary ∂U (or an empirically detected saddle/exit). Then:

(2.28) $\Delta\hat{\Phi} = \hat{\Phi}(\text{exit}) - \hat{\Phi}(\text{attractor})$

Large $\Delta\hat{\Phi}$ means deep basin (strong G-like stabilization).

G3. Contraction / return tendency
Estimate the symmetric part of the local Jacobian of $\hat{b}_G$:

(2.29) $J_G(x) = \partial\hat{b}_G/\partial x$, $C_G = \mathbb{E}[\, -\lambda_{\max}(\, (J_G + J_G^{\mathsf{T}})/2\,)\,]$

$C_G > 0$ indicates net contraction (returning dynamics).

E-like signatures (circulation / loop current / “gauge-like” behavior)

E1. Probability-current magnitude
Using $\hat{j}$:

(2.30) $M_E = \mathbb{E}[\, \|\hat{j}(x)\| \,]$

Nonzero M_E is the cleanest “there is circulation” indicator.

E2. Circulation along an observed loop
For an empirically detected loop trajectory $\Gamma = \{x[0],\ldots,x[N]\}$, define the discrete line integral:

(2.31) $\mathrm{Cir}\hat{}_E(\Gamma) = \Sigma_{n=0}^{N-1}\, \hat{b}_E(x[n]) \cdot (\, x[n+1] - x[n] \,)$

If $\mathrm{Cir}\hat{}_E$ is consistently nonzero across cycles, the loop is genuinely E-like rather than a noisy G-like orbit.

E3. Curl / rotational intensity (when $\mathcal{M} \subseteq \mathbb{R}^3$ or local charts exist)
In coordinates where curl is defined:

(2.32) $R_E = \mathbb{E}[\, \|\nabla \times \hat{b}_E(x)\| \,]$

W-like signatures (switching / triggers / discrete regime changes)

W1. Event rate and waiting-time law
Let $\{t_k\}$ be detected jump times. Define:

(2.33) $r_W = (\#\text{ of jumps in window } T) / T$

and estimate the waiting-time survival:

(2.34) $\hat{S}(\tau) = \mathbb{P}(\, t_{k+1} - t_k > \tau \,)$

Heavy tails or strong state-dependence of $\hat{\lambda}(x)$ indicate W-dominance.

W2. Regime discontinuity score (distributional shift)
Let pre- and post-event transition statistics define two empirical kernels $\hat{P}_{pre}$ and $\hat{P}_{post}$; then:

(2.35) $D_W = \mathrm{KL}(\, \hat{P}_{post} \,\|\, \hat{P}_{pre} \,)$

Large D_W indicates “true switch” rather than a large but continuous fluctuation.

S-like signatures (confinement / constraints / lock-in topology)

S1. Constraint residual smallness (closure)
If $\hat{g}$ is available:

(2.36) $R_S = \mathbb{E}[\, \|\hat{g}(x)\|^2 \,]$

Small R_S indicates states remain close to $\mathbb{N}$ (tight confinement).

S2. Strong normal restoring flow
Let $\hat{n}(x)$ be the estimated normal direction (columns of $J_{\hat{g}}^{\mathsf{T}}$ spanning normal space). Define normal drift magnitude:

(2.37) N_S = $\mathbb{E}[\ \|\hat{P}_{\perp}(x)\,\hat{b}(x)\|\ ]$

Large N_S indicates strong closure forces pushing back toward $\mathbb{N}$.

S3. Effective dimension reduction (manifold evidence)
Let $\lambda_1 \geq \ldots \geq \lambda_d$ be eigenvalues of the covariance of x in a loop. Define:

(2.38) $d_{eff}(\eta) = \min\{\ k : (\Sigma_{i=1}^{k} \lambda_i)/(\Sigma_{i=1}^{d} \lambda_i) \geq \eta\ \}$

Small d_eff at high η (e.g., η=0.95) indicates confinement to a low-dimensional set.

S4. Topological signature (when feasible)
If $\mathbb{N}$ is reconstructed, record invariants (connected components, holes) as a discrete label:

(2.39) Top_S = (#components($\mathbb{N}$), $\beta_1(\mathbb{N})$, $\beta_2(\mathbb{N})$, …)

(β_i are Betti numbers; if too heavy, start with #components only—already powerful for "membership/contract" style lock-in.)

2.5 A single "force dominance" summary (optional but marketing-useful)

For a loop L (identified in Section 1), define energy-like shares:

(2.40) $D_G = \mathbb{E}_L[\ \|\hat{b}_G\|^2\ ] / \mathbb{E}_L[\ \|\hat{b}_\|\|^2\ ]$

(2.41) $D_E = \mathbb{E}_L[\ \|\hat{b}_E\|^2\ ] / \mathbb{E}_L[\ \|\hat{b}_\|\|^2\ ]$

(2.42) $D_S = \mathbb{E}_L[\ \|\hat{b}_S\|^2\ ] / \mathbb{E}_L[\ \|\hat{b}\|^2\ ]$

(2.43) $D_W = (1/T)\ \Sigma_k\ \|\Delta x_k\|^2$ (jump energy rate over window T)

These D_* are not "truth"; they are **operational tags** used to choose which operator channel (Pump/Probe/Switch/Couple) will be most informative in the Minimal Experiment Protocol later.

---

3. Loop existence and "objecthood" (why Ξ is even meaningful)

Protocol 2 treats "objects" as **loop-bearing open systems**: if a boundary+protocol produces **recurrent closure** that is (i) stable enough to be identifiable, and (ii) not leaking away too fast, then the system supports a **low-dimensional effective description** Ξ=(ρ,γ,τ). This section makes that precise.

3.1 Loop criterion = recurrence + return-map stability + bounded leakage

Fix a protocol P=(B,Δ,h,u) as in Section 1, and define the sampled observable z[n]=h(x(nΔ)).

We define a *candidate loop tube* $R \subseteq \mathcal{Z}$ (a region in observable space), and a *Poincaré-like section* $\Sigma \subseteq R$ (a codimension-1 "crossing surface" inside the tube, in whichever coordinates your

protocol uses).

3.1.1 Recurrence (you come back, repeatedly, on a finite timescale)

Define the (discrete-time) return time to R:

(3.1) $\tau_R = \inf\{ n \geq 1 : z[n] \in R \mid z[0] \in R \}$

Recurrence requirement:

(3.2) $\mathbb{P}(\tau_R < \infty \mid z[0] \in R) = 1$

Finite-mean requirement (prevents "once in a billion years"):

(3.3) $\mathbb{E}[\tau_R \mid z[0] \in R] \leq T_{rec}$

Here T_{rec} is a declared tolerance (protocol-level constant).

Practical estimator from logs: if you have M observed returns with return lengths $\{\tau_R^{(k)}\}$, then

(3.4) $\hat{T}_{rec} = (1/M) \Sigma_{k=1}^{M} \tau_R^{(k)}$

and you accept recurrence if $\hat{T}_{rec}$ is stable across sliding windows.

3.1.2 Return map stability (the loop has a stable closure law)

Choose a section Σ with an orientation rule (e.g., "crossing in the + direction"). Let s_k be the k-th intersection point on Σ (in section coordinates).

Define the empirical return map:

(3.5) $s_{k+1} = F(s_k) + \varepsilon_k$

A loop is *map-stable* if both a residual bound and a stability bound hold.

Residual smallness:

(3.6) $\mathbb{E}[\|\varepsilon_k\|^2] \leq \varepsilon_F^2$

Local Lipschitz stability (prevents explosive divergence around the loop):

(3.7) $\|F(s) - F(s')\| \leq L_F \|s - s'\|$ for s,s' in the observed support

and we require:

(3.8) $L_F \leq 1 + \delta_F$

Interpretation:

- $L_F < 1$ is a contraction (very stable loop),
- $L_F \approx 1$ is neutrally stable (common in oscillators),
- $L_F > 1+\delta_F$ indicates the "loop" is not robust (it is a transient passage or sampling artifact).

Window invariance (the loop law should not drift wildly): for two time windows w and

w′,

(3.9) sup_{s in support} ‖F̂_w(s) − F̂_{w′}(s)‖ ≤ ε_drift

This is the operational “no proxy drift” version of map stability.

3.1.3 Bounded leakage (the object does not dissolve through the boundary)

Loops are only meaningful if the trajectory stays inside the loop tube R for many cycles most of the time.

Define the exit time from R:

(3.10) τ_exit = inf{ n ≥ 1 : z[n] ∉ R | z[0] ∈ R }

Define the per-step leakage probability:

(3.11) p_leak = ℙ(z[n+1] ∉ R | z[n] ∈ R)

and the leakage rate:

(3.12) ℓ_R = p_leak / Δ

A **bounded leakage** condition is:

(3.13) ℓ_R ≤ ℓ_max

Equivalent “per-cycle survival” form (often clearer to readers): if the loop has nominal period T (in samples), define survival over m cycles:

(3.14) S(m) = ℙ(τ_exit > mT | z[0] ∈ R)

and require:

(3.15) S(m₀) ≥ 1 − ε_surv for some declared m₀ (e.g., m₀=10 cycles)

This is the minimal “object stays intact long enough to study” condition.

3.1.4 The loop existence criterion (one line)

A loop L exists under protocol P if there exist (R,Σ,T) such that:

(3.16) Recurrence (3.2–3.3) ∧ Map-stability (3.6–3.9) ∧ Leakage bound (3.13 or 3.15)

When (3.16) holds, we treat the loop as an **operational object**: it supports repeatable behavior and admits controlled perturbation analysis.

3.2 Why this implies “objecthood” (and why Ξ is meaningful)

Once a loop satisfies (3.16), two things become true:

1. **There is a persistent closure law** (return map F is stable enough to identify).

2. **There is a persistent timescale** (recurrence time T_rec and cycle time T are stable enough to estimate).

These are exactly the prerequisites for defining **effective slow coordinates** (ρ,γ,τ) without them degenerating into "pure narrative."

A minimal mathematical way to state this is:

- the loop defines a quasi-invariant "loop measure" μ_L on $\mathcal{Z}$:

(3.17) μ_L(A) = lim_{N→∞} (1/N) Σ_{n=0}^{N−1} 1_A(z[n]) conditioned on staying in R

- and the stability conditions ensure μ_L is approximately stationary across windows (no collapse into pure drift).

Then Ξ is meaningful because it is compiled from **stable loop-level observables** (basin depth proxies, exit-cost proxies, and recovery/switching times) rather than raw microstate noise.

3.3 Multiple loops can coexist (and usually do)

Real systems are rarely single-loop. Noise, constraints, and triggers often create **multiple recurrent closures**:

- multiple basins (G-like multiplicity),
- multiple circulating modes (E-like multiplicity),
- multiple constraint-locked submanifolds (S-like multiplicity),
- multiple regimes connected by switching (W-like multiplicity).

3.3.1 Coexistence definition via time occupancy

Suppose you have K candidate loop tubes {R_i} with corresponding loops L_i satisfying (3.16).

Define long-run occupancy:

(3.18) π_i = lim_{N→∞} (1/N) Σ_{n=0}^{N−1} 1_{R_i}(z[n])

We say multiple loops **coexist** if at least two have nontrivial occupancy:

(3.19) ∃ i≠j such that π_i > 0 and π_j > 0

This captures the common reality: the system spends time in different loop modes.

3.3.2 Switching between loops (W-like edges)

Define the next-loop map: when the trajectory exits R_i, it may enter some R_j. Define the empirical transition probability:

(3.20) p_{j←i} = ℙ(next entry is into R_j | current exit is from R_i)

In data terms:

(3.21) p̂_{j←i} = (# exits from R_i followed by entry into R_j) / (# exits from R_i)

This induces a **loop transition graph** $\mathcal{G}$ with nodes {L_i} and directed edges i→j when $\hat{p}_{j \leftarrow i}$ is non-negligible.

This graph is the operational place where "multiple loops + switching" becomes a finite object you can reason about.

3.4 Basis loops: a minimal spanning set

Protocol 2 wants a **minimal set of loops** that can represent the system's observed loop behaviors without redundancy. There are two complementary (and both useful) definitions of "basis loops":

3.4.1 Basis notion A (ergodic / operational reachability basis)

Compute the directed graph $\mathcal{G}$ from (3.21). Consider its **closed communicating classes** (recurrent strongly connected components with no outgoing edges). Let these classes be $\{\mathcal{C}_1,\ldots,\mathcal{C}_M\}$.

A minimal reachability basis chooses one representative loop from each closed class:

(3.22) $\boldsymbol{\mathcal{B}}_{recur} = \{ L_{r(1)}, \ldots, L_{r(M)} \}$ where $L_{r(m)} \in \mathcal{C}_m$

Why this is a "spanning set" in practice:

- any loop outside closed classes is transient (it eventually feeds into some closed class),
- closed classes represent persistent operating modes of the object under the protocol.

This basis is minimal in the sense that removing any representative would leave at least one persistent mode uncovered.

3.4.2 Basis notion B (signature-span basis for engineering compression)

Assign each loop a **signature vector** $s_i \in \mathbb{R}^p$ built from estimable signatures (Section 2) and basic loop statistics (Section 3), e.g.:

(3.23) $s_i = (D_G, D_E, D_W, D_S, T_{rec}, \ell_R, \ldots)$

We define a basis set $\boldsymbol{\mathcal{B}}_{sig}$ as a minimal subset of loops whose signature vectors span the others within tolerance ε in a nonnegative mixture sense:

(3.24) for each i, $\min_{\alpha_{ij}\geq 0} \| s_i - \Sigma_{j\in\boldsymbol{\mathcal{B}}_{sig}} \alpha_{ij} s_j \| \leq \varepsilon$

This "nonnegative span" is intentionally chosen because:

- it matches operational mixing (spending time in different loops, convex-like composition),
- it produces interpretable decompositions ("this loop is 0.7×A + 0.3×B in signatures").

In practice, $\boldsymbol{\mathcal{B}}_{sig}$ can be found by a greedy independence rule (pick the loop that most increases span coverage) or by an NMF-style factorization on the signature matrix.

3.5 What you keep (so we don't lose the logic later)

This section's spine can be summarized as:

- Loop existence is an **operational triad**:

(3.25) Loop exists ⇔ recurrence ∧ return-map stability ∧ bounded leakage

- Multiple loops coexist and induce a **finite loop graph** with transition probabilities:

(3.26) $\mathcal{G}$: nodes = loops, edges weighted by $\hat{p}_{j \leftarrow i}$

- Basis loops can mean either:
  - minimal representatives of persistent recurrent classes (3.22), or
  - minimal signature-spanning set for compression (3.24).

---

4. Compression step: compiling Ξ=(ρ,γ,τ) from loop-level observables

Once a loop **L** exists (Section 3), we can compress its behavior into a small coordinate triple:

(4.1) $\Xi(L) = (\rho(L), \gamma(L), \tau(L))$

This is a **protocol-dependent**, **operational** coordinate system: ρ,γ,τ are *compiled from measurable loop observables* and must pass stability gates.

4.0 Preliminaries: what data a "loop" gives you

Fix a loop tube $R \subseteq \mathcal{Z}$ and a sampling step Δ. From a long run of $z[n]=h(x(n\Delta))$ you can estimate:

- **Occupancy / stationary mass inside the loop tube**:
  (4.2) $\hat{\pi}_R = (1/N) \Sigma_{n=0}^{N-1} 1_{R}(z[n])$

- **Exit events and leakage rate** (Section 3):
  (4.3) $\tau_{exit} = \inf\{ n \geq 1 : z[n] \notin R \mid z[0] \in R \}$
  (4.4) $\hat{\ell}_R = 1/(\Delta \cdot \mathbb{E}[\tau_{exit}])$ (rate form)

- **Return times** (recurrence):
  (4.5) $\tau_R = \inf\{ n \geq 1 : z[n] \in R \mid z[0] \in R \}$
  (4.6) $\hat{T}_{rec} = \Delta \cdot \mathbb{E}[\tau_R]$ (time units)

- **Return map** on a section Σ (Section 3):
  (4.7) $s_{k+1} = \hat{F}(s_k) + \varepsilon_k$
  (4.8) $\text{residual}_F = \mathbb{E}[\|\varepsilon_k\|^2]$
  (4.9) $\hat{L}_F$ = empirical Lipschitz / slope proxy of $\hat{F}$

- **Jump/switch statistics** (Section 2 W-like):
  (4.10) $\hat{r}_W$ = (# detected jump events)/T_{obs}
  (4.11) $\hat{\tau}_{jump} = 1/\hat{r}_W$ (mean time between detected switches)

These are “loop-level observables.” Now we compile ρ,γ,τ.

4.1 ρ: stock / saturation / potential depth proxy (G-dominant sensitivity)

ρ is the “how much is accumulated / how deep is the basin” coordinate. It should respond strongly to **G-like** changes (potential/depth) and to **Pump-like** interventions later.

Definition 4.1 (Canonical ρ from an effective potential depth)

If you can estimate an effective stationary density $\hat{\pi}(z)$ on observable space (or on an embedding of it), define an effective potential:

(4.12) $\hat{\Phi}(z) = -\log(\hat{\pi}(z) + \varepsilon_0)$ (defined up to an additive constant)

Choose a loop “core” region $C \subseteq R$ (e.g., the highest-density subset of R), and define a loop boundary shell ∂R (or an exit surface). Then:

(4.13) $\hat{\rho}_{\Phi}(L) = \text{median}_{z\in\partial R}\, \hat{\Phi}(z) - \text{median}_{z\in C}\, \hat{\Phi}(z)$

Interpretation: large $\hat{\rho}_{\Phi}$ means “core is much more probable than boundary,” i.e., a deep basin.

Why this is G-dominant: it is exactly a basin depth in the effective potential representation.

Proxy template A (escape-time depth proxy; no density estimation needed)

If you do not want to estimate $\hat{\pi}$, use exit time from the loop tube as a depth proxy:

(4.14) $\hat{\rho}_{exit}(L) = \log(\, \mathbb{E}[\tau_{exit}] + 1\,)$

This is extremely robust: deeper basins have longer mean exit times.

Proxy template B (stock/saturation proxy when a “stock variable” exists)

If the protocol has a measurable “stock” observable q(z) (inventory, energy, capital, resource pool, backlog), define:

(4.15) $\hat{\rho}_{stock}(L) = \mathbb{E}[\, q(z) \mid z \in R\,]$

This is the “accountant-friendly” version: ρ is literally average stored amount while in-loop.

Practical recommendation (choose one canonical ρ)

Pick one as canonical per paper / per system class:

(4.16) $\hat{\rho}(L) = \hat{\rho}_{\Phi}(L)$ if $\hat{\pi}$ is reliable
(4.17) $\hat{\rho}(L) = \hat{\rho}_{exit}(L)$ otherwise

Keep the alternatives only as cross-checks (Gate ρ-G3 below).

4.2 γ: closure / exit-cost / coupling proxy (S-dominant sensitivity)

γ is "how closed / locked-in / constraint-bound" the loop is. It should respond strongly to **S-like** changes (constraints, coupling, confinement) and later to **Couple-like** interventions.

Definition 4.2 (Canonical γ from confinement + survival)

Use two independently estimable ingredients:

1. **Survival / membership strength** via leakage:
   (4.18) $\hat{\gamma}_{surv}(L) = \log(1/\hat{\ell}_R + 1)$ (larger = less leakage)

2. **Confinement / constraint tightness** (when a constraint residual is defined):
   Assume you have a constraint function $\hat{g}(z)$ such that $\hat{g}(z)=0$ represents "in-constraint."
   Define residual and normal restoring magnitude:

(4.19) $\hat{R}_S = \mathbb{E}[\, \|\hat{g}(z)\|^2 \mid z \in R \,]$
(4.20) $\hat{N}_S = \mathbb{E}[\, \|\hat{P}_{\perp}(z)\, \hat{b}(z)\| \mid z \in R \,]$ (normal restoring strength)

Then define a dimensionless confinement index:

(4.21) $\hat{\gamma}_{conf}(L) = \log(1 + \hat{N}_S/(\sqrt{(\hat{R}_S)}+\varepsilon_0))$

Now combine:

(4.22) $\hat{\gamma}(L) = w_1\, \hat{\gamma}_{surv}(L) + w_2\, \hat{\gamma}_{conf}(L)$

Default weights (if you don't want a tuning discussion): $w_1=w_2=1$.

Proxy template A (pure survival / exit-cost; simplest and often sufficient)

If you don't have $\hat{g}$ or $\hat{P}_{\perp}$, just use survival:

(4.23) $\hat{\gamma}_{simple}(L) = \log(1/\hat{\ell}_R + 1)$

This already captures "closure" and "exit cost" operationally.

Proxy template B (coupling-sensitive closure; optional)

If you already compute loop-to-loop coupling κ later (Section 7), you may incorporate it as an *optional* correction:

(4.24) $\hat{\gamma}_{cpl}(L) = \hat{\gamma}(L) + \alpha \cdot \log(1 + \Sigma_{B\neq L} |\hat{\kappa}_{L\leftarrow B}|)$

This makes γ reflect that some loops are "closed because they are strongly bound to others."

4.3 τ: recovery / switching timescale proxy (W-

dominant sensitivity)

τ is “how fast the loop recovers and/or switches.” It should be sensitive to **W-like** switching (jump hazard, regime changes), and also reflect within-loop recovery time.

Because many systems have *both* a recovery timescale and a switching timescale, we explicitly define both and then choose a canonical τ.

Definition 4.3 (Recovery time τ_rec from return-map contraction)

From the return map s_{k+1}=F^(s_k)+ε_k, define a contraction-like factor:

(4.25) ĉ = min(1, L^_F) (a conservative stability proxy)

If ĉ<1, the number of cycles to reduce a deviation by factor η is:

(4.26) n_η = log(η)/log(ĉ)

Convert to time using the cycle period T^_cycle:

(4.27) τ^_rec(L) = T^_cycle · max(1, n_η)

If ĉ≈1 (neutral stability), use recurrence time instead:

(4.28) τ^_rec(L) = T^_rec when ĉ is near 1

Definition 4.4 (Switching time τ_sw from jump / regime changes)

Using detected jump rate r^_W:

(4.29) τ^_sw(L) = 1/r^_W (mean time between switches)

If you track loop-to-loop transitions i→j, define switching time out of loop i:

(4.30) τ^_out(i) = Δ · 𝔼[time spent in R_i before first entry to any R_j, j≠i]

In many cases τ^_sw ≈ τ^_out; keep both only if they disagree (Gate τ-G2).

Canonical τ (one scalar)

A single scalar τ should represent “the slow time that matters for control.” A safe default is:

(4.31) τ^(L) = max(τ^_rec(L), τ^_sw(L))

Interpretation: the loop’s meaningful time constant is whichever is slower—recovering within the loop, or switching out of it.

4.4 Proxy templates summary (what you can literally plug in)

For each loop L:

ρ templates

- (4.13) depth from $\hat{\Phi}$
- (4.14) log mean exit time
- (4.15) mean stock variable

γ templates

- (4.23) log survival $(1/\hat{\ell}_R)$
- (4.21) confinement ratio $\log(1 + \hat{N}_S/\sqrt{\hat{R}_S})$
- (4.24) optional coupling augmentation

τ templates

- (4.27–4.28) recovery from return map / recurrence
- (4.29–4.30) switching from hazard / loop transitions
- (4.31) canonical max

4.5 Identifiability conditions + proxy stability gates

A proxy is only acceptable if it is (i) *identifiable from available data under the protocol*, and (ii) *stable enough to be treated as a coordinate*. We enforce this with explicit gates.

Gate 0 (Loop validity prerequisite)

You may only compile Ξ if the loop passes Section 3:

(4.32) Loop valid $\Leftrightarrow$ recurrence $\wedge$ return-map stability $\wedge$ bounded leakage

If Gate 0 fails, Ξ is undefined (you must refine boundary/protocol or split loops).

Gate ρ-G1 (Window stability)

Compute $\hat{\rho}$ on K disjoint time windows. Let $\hat{\mu}_\rho$ be the mean and $\hat{s}_\rho$ the standard deviation across windows. Require:

(4.33) $CV_\rho = \hat{s}_\rho/(\hat{\mu}_\rho + \varepsilon_0) \le c_\rho$

Typical choice: c_ρ in [0.1, 0.3] depending on noise tolerance.

Gate ρ-G2 (Model-choice consistency)

If you compute two independent ρ proxies (depth vs exit-time), require they are monotone consistent across loops:

(4.34) $\text{corr_rank}(\{\hat{\rho}_\Phi(L_i)\}, \{\hat{\rho}_{exit}(L_i)\}) \ge r_{min}$

This prevents a "ρ" that is actually measuring something else.

Gate γ-G1 (Boundary accounting / survival stability)

Require leakage estimate is meaningful and stable:

(4.35) $\hat{\ell}_R \le \ell_{max}$ and $CV_\ell \le c_\ell$

If leakage is too large or too unstable, "closure" is not well-defined for that boundary.

Gate γ-G2 (Confinement evidence, if using $\hat{\gamma}_{conf}$)

If you use constraint-based $\hat{\gamma}_{conf}$, you must actually be confined:

(4.36) $d_{eff}(0.95) \le d_{max}$ and/or $\hat{R}_S$ is small and stable

Otherwise the "constraint manifold" is not a real property of the loop under this protocol.

Gate τ-G1 (Timescale separation for meaningful τ)

To treat τ as a loop-level time constant, you need enough cycles/events:

(4.37) $N_{cycles} \ge N_{min}$ and $N_{jumps} \ge J_{min}$ (if using $\hat{\tau}_{sw}$)

If not, $\hat{\tau}$ becomes a sampling artifact.

Gate τ-G2 (Switching vs recovery coherence)

If both $\hat{\tau}_{rec}$ and $\hat{\tau}_{sw}$ are computed, require they are stable and not contradictory in a way indicating misclassification:

(4.38) $\hat{\tau}_{sw} \ll \hat{\tau}_{rec}$ with frequent exits but "stable return map" ⇒ likely loop segmentation error
(4.39) $\hat{\tau}_{sw} \gg \hat{\tau}_{rec}$ with near-zero jumps ⇒ W-dominance absent; treat τ as recovery-only

This gate tells you whether your loop graph is correctly constructed.

Gate Ξ-G1 (Intervention invariance under "null probes")

Apply a tiny "null" perturbation that should not materially change the loop (e.g., a very small Probe that does not alter boundary or constraints). Require:

(4.40) $\|\hat{\Xi}_{after} - \hat{\Xi}_{before}\| \le \varepsilon_\Xi$

If this fails, your proxies are too observer-sensitive; either change h, change Δ, or strengthen the loop definition.

4.6 Output artifact: the Ξ-Card for a loop

For each accepted loop L, record:

(4.41) $\hat{\Xi}(L) = (\hat{\rho}(L), \hat{\gamma}(L), \hat{\tau}(L))$ with gates passed and proxy choices declared

That “with proxy choices declared” is not bureaucracy—it’s what prevents future readers from mistaking Ξ for metaphysics.

---

5. Operator grammar as controlled perturbations of the generator

Protocol 2 treats “control” as **generator perturbation**. The four operators are not metaphors; they are four distinct **knob families** that deform $\mathscr{L}$ in different, estimable ways.

We write the controlled generator as:

(5.1) $\mathscr{L}_u = \mathscr{L}_0 + u_P \mathscr{L}_P + u_Q \mathscr{L}_Q + u_S \mathscr{L}_{Sw} + u_C \mathscr{L}_C + \mathscr{L}_{res}(u)$

where:

- $\mathscr{L}_0$ is the baseline generator under the fixed protocol,
- $u = (u_P, u_Q, u_S, u_C)$ are scalar (or low-dimensional) control inputs,
- $\mathscr{L}_{res}(u)$ collects higher-order terms, nonlinearities, and unmodeled interactions (tracked, not hidden).

The “marketing hook” (G↔Pump, E↔Probe, W↔Switch, S↔Couple) becomes **precise** once each $\mathscr{L}_*$ is defined at generator level.

5.1 Preliminaries: controlled Markov dynamics and identifiability goal

Assume that under control $u(t)$, $x(t)$ remains Markov and its generator is well-defined. The only thing we require is local identifiability:

(5.2) $(\mathscr{L}_u - \mathscr{L}_0)\,\varphi$ is estimable from (trajectory logs + known $u(t)$) for test functions φ

In practice this means we will later estimate **gain maps**:

(5.3) $G_\Xi = \partial\Xi/\partial u$ (local, loop-specific)

but Section 5 focuses on defining the perturbation channels.

5.2 Operator 1: Pump (P) = source/sink and potential-depth knob (Gravity-like)

Definition 5.1 (Pump as potential deformation and/or injection–drain)

Pump is the operator family that changes the **effective potential depth** and therefore strongly controls ρ.

We define Pump as a deformation of the drift by a gradient term:

(5.4) $b_u(x) = b_0(x) - u_P \nabla\Psi_P(x)$

equivalently:

(5.5) $(\mathscr{L}_P \varphi)(x) = (-\nabla \Psi_P(x)) \cdot \nabla \varphi(x)$

Here Ψ_P is a declared "pump potential" (design choice, not discovered truth). Typical choices:

- Ψ_P pushes states toward an attractor core (deepening basin),
- or away from it (shallowing basin),
- or injects/dissipates a stock variable.

Canonical pairing: Pump is **gravity-like** because it changes the gradient/potential part of the generator (G component). This is exact at the level of Eq. (5.4–5.5), not a metaphor.

Minimal signature of Pump effectiveness

Under small u_P, Pump should move ρ most:

(5.6) $|\partial \rho / \partial u_P| \gg |\partial \rho / \partial u_Q|, |\partial \rho / \partial u_S|, |\partial \rho / \partial u_C|$ (loop-local dominance expectation)

(This is an empirical check, not an axiom.)

5.3 Operator 2: Probe (Q) = measurement/interface knob and circulation alignment (EM-like)

Probe is the most misunderstood operator, so we define it carefully.

Probe is the operator family that changes the **observer-interface channel**: what is coupled, what is measured, what is "lit up," and therefore can induce backreaction. In Protocol 2, Probe is the precise handle for **E-like** (circulation/gauge-like) phenomena because it can change phase alignment and induce currents without changing basin depth.

Definition 5.2 (Probe as controlled coupling to an auxiliary channel)

Introduce an auxiliary probe process $y(t)$ (sensor, interface, messaging channel, measurement apparatus) and a coupling Hamiltonian-like term $H_Q(x,y)$ or coupling drift term $C_Q(x,y)$. The joint generator on (x,y) is:

(5.7) $\mathscr{L}_u^{xy} = \mathscr{L}_0^x + \mathscr{L}_0^y + u_Q \mathscr{L}_Q^{xy}$

The induced marginal effect on x is captured by an effective generator perturbation:

(5.8) $\mathscr{L}_u^x \approx \mathscr{L}_0^x + u_Q \mathscr{L}_Q^x$ (after eliminating y under the protocol)

You can implement $\mathscr{L}_Q^x$ as a drift perturbation that is **approximately divergence-free under the loop measure**, i.e., it mostly changes current:

(5.9) $b_u(x) = b_0(x) + u_Q b_Q(x)$, with $\nabla \cdot (b_Q(x) \pi_L(x)) \approx 0$

Equivalently, it increases the probability current magnitude M_E (Section 2) without systematically deepening/shallowing Φ.

Canonical pairing: Probe is **EM-like** because its defining signature is *current/circulation control* (Eq. 5.9), and because it is the channel through which observer coupling and gauge-like degrees of freedom enter.

Minimal signature of Probe effectiveness

Probe should change E-like scores (current/circulation) more than it changes potential depth:

(5.10) $|\partial M_E/\partial u_Q| \gg |\partial \Delta\Phi/\partial u_Q|$

5.4 Operator 3: Switch (Sw) = jump kernel / trigger knob (Weak-like)

Switch is the operator family that changes **rare events and regime transitions**. In the generator, that is exactly the jump operator.

Definition 5.3 (Switch as controlled hazard and jump destination kernel)

Recall the jump generator:

(5.11) $(\mathscr{L}_{\text{jump}}\varphi)(x) = \int_{\{\mathcal{Y}\}} [\varphi(x+\kappa(x,y)) - \varphi(x)]\, \nu(x,dy)$

Define a switch control u_S that modifies either:

- the hazard intensity $\lambda(x)$, or
- the jump destination distribution $K(x,dy)$, or
- the jump size $\kappa(x,y)$.

A canonical parameterization is:

(5.12) $\nu_u(x,dy) = \lambda_u(x)\, K_u(x,dy)$

(5.13) $\lambda_u(x) = \lambda_0(x) \exp(u_S\, s(x))$

(5.14) $K_u(x,dy) \propto K_0(x,dy) \exp(u_S\, r(x,y))$

Then:

(5.15) $(\mathscr{L}_{Sw}\varphi)(x) = \partial/\partial u_S \left[\int (\varphi(x+\kappa) - \varphi(x))\, \nu_u(x,dy) \right]$ at $u_S=0$

Canonical pairing: Switch is **weak-like** because it governs the *trigger channel*—small continuous changes in state or control can produce a discrete mode change via ν_u.

Minimal signature of Switch effectiveness

Switch should primarily change τ through the switching time:

(5.16) $\partial\tau_{sw}/\partial u_S \neq 0$ and typically $|\partial\tau_{sw}/\partial u_S| \gg |\partial\tau_{rec}/\partial u_S|$

5.5 Operator 4: Couple (C) = closure / constraint / binding knob (Strong-like)

Couple is the operator family that changes **closure strength**, **exit cost**, and **binding topology**. It is S-like because it increases confinement and reduces leakage.

Definition 5.4 (Couple as constraint penalty / binding energy scaling)

Let $\mathbb{N}$ be a constraint manifold $g(x)=0$ (Section 2). Define a coupling/closure penalty:

(5.17) $\Phi_C(x) = (1/2)\,\|g(x)\|^2$

Couple scales the penalty strength:

(5.18) $b_u(x) = b_0(x) - u_C\,\nabla\Phi_C(x)$

so:

(5.19) $(\mathscr{L}_C\,\varphi)(x) = (-\nabla\Phi_C(x))\cdot\nabla\varphi(x)$

As u_C increases, deviations from $\mathbb{N}$ become increasingly costly, and the loop becomes more "locked in."

Couple can also bind two subsystems x_A and x_B:

(5.20) $g_{AB}(x_A,x_B) = 0$ (a relational constraint)

(5.21) $\Phi_C(x_A,x_B) = (1/2)\,\|g_{AB}(x_A,x_B)\|^2$

This is the clean generator-level representation of "strong binding" between loops/objects.

Canonical pairing: Couple is **strong-like** because it implements confinement and binding by increasing penalty strength and reducing degrees of freedom (Eq. 5.17–5.21).

Minimal signature of Couple effectiveness

Couple should strongly increase γ and reduce leakage:

(5.22) $\partial\gamma/\partial u_C > 0$ and $\partial\ell_R/\partial u_C < 0$ (expected signs)

5.6 Canonical pairing (precisely stated)

The canonical pairing is now a theorem-like statement about which generator parts are perturbed:

(5.23) Pump perturbs $\mathscr{L}_G$ primarily: $\mathscr{L}_P \subset$ gradient/potential deformation

(5.24) Probe perturbs $\mathscr{L}_E$ primarily: $\mathscr{L}_Q \subset$ current/circulation / interface coupling

(5.25) Switch perturbs $\mathscr{L}_W$ primarily: $\mathscr{L}_{Sw} \subset$ jump kernel $\nu(x,dy)$

(5.26) Couple perturbs $\mathscr{L}_S$ primarily: $\mathscr{L}_C \subset$ constraint penalty / binding closure

Marketing line that remains mathematically honest:
"G/E/W/S are not philosophical labels; they are four estimable generator mechanisms. Pump/Probe/Switch/Couple are four controlled perturbation families that target those mechanisms."

5.7 Practical note: distinctness and cross-talk (so readers don't over-idealize)

In real systems the channels are not perfectly orthogonal. We model cross-talk explicitly by allowing:

(5.27) $\mathscr{L}_u = \mathscr{L}_0 + \Sigma_{i\in\{P,Q,Sw,C\}} u_i \mathscr{L}_i + \Sigma_{i<j} u_i u_j \mathscr{L}_{ij} + \dots$

The “canonical pairing” means:

- the **first-order** effect of each operator is most identifiable in its paired archetype,
- higher-order terms are tracked and tested (not denied).

---

6. Minimal Experiment Protocol (MEP): how to get a first usable gain artifact

This section gives a **first-pass, reproducible** procedure to estimate a local gain map:

(6.1) $\Delta\Xi \approx G\,\Delta u$

with explicit **regime-rejection gates** (jump / KL) so you don’t accidentally fit a linear gain across a mode switch.

6.1 Setup: what MEP assumes you already have

From Sections 1–4 you have, for a chosen loop tube R and protocol P:

- a validated loop L (recurrence + stable return map + bounded leakage),
- compiled proxies $\Xi\hat{}(L)=(\rho\hat{},\gamma\hat{},\tau\hat{})$,
- operator channels $u = (u_P,u_Q,u_{Sw},u_C)$ (Pump, Probe, Switch, Couple),
- time series logs $z[n]$ (or $x[n]$) sampled at step Δ.

MEP outputs:

- an estimated **gain matrix** $\hat{G} \in \mathbb{R}^{3\times4}$ mapping $\Delta u \to \Delta\Xi$,
- a **loop tag** (primary archetype: G/E/W/S, plus confidence).

6.2 One-channel perturbation design (keep it orthogonal on purpose)

MEP uses **one-channel perturbations** so gains are identifiable even in messy systems.

Let u^0 be the baseline operator setting (often $u^0=0$). For channel $i \in \{P,Q,Sw,C\}$, define a perturbation amplitude δu_i and hold all others fixed:

(6.2) $u(t) = u^0 + \delta u_i\, e_i$ for t in perturb window, and $u(t)=u^0$ otherwise

where e_i is the i-th standard basis vector.

You run **three windows** (each long enough to compute $\hat{\Xi}$ stably):

- W0: baseline window
- W1: perturbed window (only one channel changed)
- W2: recovery window (back to baseline)

Let the window durations be T0,T1,T2.

6.3 Measuring ΔΞ (window-averaged, gate-aware)

Compute $\hat{\Xi}$ on each window using the same proxy choices and gates from Section 4:

(6.3) $\hat{\Xi}_0 = \hat{\Xi}(W0)$, $\hat{\Xi}_1 = \hat{\Xi}(W1)$, $\hat{\Xi}_2 = \hat{\Xi}(W2)$

Define the naive perturbation effect:

(6.4) $\Delta\hat{\Xi}_{naive} = \hat{\Xi}_1 - \hat{\Xi}_0$

and a recovery sanity check:

(6.5) $\Delta\hat{\Xi}_{recov} = \hat{\Xi}_2 - \hat{\Xi}_0$

A **basic linear-response sanity condition** is that recovery is small compared to the perturbation effect:

(6.6) $\|\Delta\hat{\Xi}_{recov}\| \le \varepsilon_{rec} \cdot \|\Delta\hat{\Xi}_{naive}\| + \varepsilon_0$

If (6.6) fails, you likely have drift, boundary leakage, or a regime change; do not accept a linear gain from this trial.

6.4 Regime rejection gates: jump gate and KL gate

MEP rejects (or re-labels) trials where the perturbation caused a **mode switch** rather than a small local deformation.

6.4.1 Jump gate (direct discontinuity detection)

Using z[n] (or x[n]), define increments:

(6.7) $\Delta z[n] = z[n+1] - z[n]$

Define a jump indicator with a declared threshold θ:

(6.8) $J[n] = 1\{ \|\Delta z[n]\| > \theta \}$

Compute the jump rate in each window:

(6.9) $\hat{r}_J(Wk) = (1/((Nk-1)\Delta)) \Sigma_{n\in Wk} J[n]$

Declare **jump-triggered regime change** if:

(6.10) $\hat{r}_J(W1) - \hat{r}_J(W0) \ge r_{J,min}$

If (6.10) triggers, you **do not** fit a local linear gain across W1 as “same regime.” Instead you either:

- treat this trial as **Switch-dominant evidence** (good for tagging), or
- segment W1 into pre-jump part and fit only pre-jump samples (if enough remain).

6.4.2 KL gate (distributional shift without obvious jumps)

Some regime changes are “soft” (no single massive jump) but still invalidate linearization. We detect them via distribution shift of increments.

Let $\hat{P}_k$ be the empirical distribution of increments Δz in window W_k (e.g., histogram / KDE in a chosen embedding). Define:

(6.11) $D_{KL}(W_1||W_0) = KL(\hat{P}_1 \| \hat{P}_0)$

Use a symmetric form if you prefer stability:

(6.12) $D_{sym} = 1/2\ KL(\hat{P}_1\|\hat{P}_0) + 1/2\ KL(\hat{P}_0\|\hat{P}_1)$

Declare **regime change** if:

(6.13) $D_{sym} \geq \kappa_{KL}$

Interpretation:

- KL small ⇒ W1 is a local perturbation in the same operating mode ⇒ linear gain plausible.
- KL large ⇒ you crossed into a different mode (often W-like) ⇒ do not fit a single local gain across it.

Important: KL is computed on increments because increments encode generator changes more directly than raw state occupancy.

6.5 Fitting a local gain map (first usable version)

6.5.1 Local linear response model

Define the local gain matrix:

(6.14) $\Delta\Xi = G\,\Delta u + \varepsilon$

where:

- $\Delta u \in \mathbb{R}^4$ is the operator change vector,
- $\Delta\Xi \in \mathbb{R}^3$ is the resulting change in (ρ,γ,τ),
- $G \in \mathbb{R}^{\{3\times4\}}$ is the loop-local gain matrix.

MEP estimates G by combining one-channel trials.

6.5.2 One-channel gain estimate (robust and minimal)

If you perturb only channel i, then $\Delta u = \delta u_i e_i$ and the gain column g_i is:

(6.15) $\hat{g}_i = \Delta\hat{\Xi} / \delta u_i$ (a 3×1 vector)

But use the gate-cleaned estimate:

- if Jump/KL gates do not trigger, set $\Delta\hat{\Xi} = \hat{\Xi}_1 - \hat{\Xi}_0$
- if gates trigger, either reject, or fit on pre-switch segment only.

Stack the columns:

(6.16) $\hat{G} = [\ \hat{g}_P\ \hat{g}_Q\ \hat{g}_{Sw}\ \hat{g}_C\ ] \in \mathbb{R}^{3\times4}$

This is your **first usable gain artifact**.

6.5.3 Minimal uncertainty reporting (so the gain is not "fake precise")

Compute $\hat{\Xi}$ in sub-blocks inside each window (e.g., split W0 and W1 into B blocks). Let $\hat{\Xi}_{0,b}$ and $\hat{\Xi}_{1,b}$ be block estimates. Then:

(6.17) $\Delta\hat{\Xi}_b = \hat{\Xi}_{1,b} - \hat{\Xi}_{0,b}$

(6.18) $\hat{g}_{i,b} = \Delta\hat{\Xi}_b / \delta u_i$

Report mean and dispersion:

(6.19) $\hat{g}_i = \text{mean}_b(\hat{g}_{i,b})$, $SE(\hat{g}_i)=\text{std}_b(\hat{g}_{i,b})/\sqrt{B}$

Even if you don't run a full statistics section, Eq. (6.19) gives readers a clear "confidence smell test."

6.6 Output: loop tag (primary archetype) from *both* structure and gain

Protocol 2 tags loops using two independent sources:

1. **Structure signatures** (Section 2): D_G,D_E,D_W,D_S (or your chosen signature scores).
2. **Gain dominance** from MEP: which operator channel most strongly moves the "paired coordinate."

6.6.1 Gain-based dominance scores (paired-coordinate rule)

Define paired sensitivities:

(6.20) $S_P = |\partial\rho/\partial u_P| \approx |(\hat{g}_P)_\rho|$

(6.21) $S_C = |\partial\gamma/\partial u_C| \approx |(\hat{g}_C)_\gamma|$

(6.22) $S_{Sw} = |\partial\tau/\partial u_{Sw}| \approx |(\hat{g}_{Sw})_\tau|$

Probe is about circulation/current more than Ξ directly, so if you also logged an E-signature (e.g., circulation $\widehat{Cir}_E$ or current magnitude M_E), define:

(6.23) $S_Q = |\partial M_E / \partial u_Q|$ (preferred)
If you don't have M_E, use a fallback:
(6.24) $S_Q = \|\hat{g}_Q\|$ (weak fallback)

Normalize:

(6.25) $\hat{S}_i = S_i / (S_P + S_Q + S_{Sw} + S_C + \varepsilon_0)$

Gain-based primary operator:

(6.26) $i^* = \text{argmax}_i \hat{S}_i$

Map operator → archetype by the canonical pairing:

(6.27) Pump→G , Probe→E , Switch→W , Couple→S

So the **gain-based tag** is:

(6.28) $\text{Tag}_{\text{gain}}(L) = \text{map}(i^*)$

6.6.2 Combined tag (structure + gain)

Let structure-based normalized scores be:

(6.29) $\tilde{D}_A = D_A / (D_G + D_E + D_W + D_S + \varepsilon_0)$ for $A \in \{G,E,W,S\}$

Define gain-based archetype scores $\tilde{G}_A$ by mapping $\hat{S}_i$ into G/E/W/S via (6.27).

Then define a combined score:

(6.30) $\text{Score}_A = \alpha \tilde{D}_A + (1-\alpha) \tilde{G}_A$, $\alpha \in [0,1]$

Primary loop tag:

(6.31) $\text{Tag}(L) = \text{argmax}_A \text{Score}_A$

And a simple confidence margin:

(6.32) $\text{Conf}(L) = \text{Score}_{\max} - \text{Score}_{2\text{nd}}$

This produces a tag that is:

- **not** purely narrative (it is measured),
- **not** purely intervention-based (it respects baseline structure),
- and remains readable as a "marketing artifact."

6.7 The Gain Card (the portable artifact you can paste into reports)

For each loop L, MEP outputs:

(6.33) $\hat{\Xi}(L) = (\hat{\rho}, \hat{\gamma}, \hat{\tau})$

(6.34) $\hat{G}(L) = [\ \hat{g}_P\ \hat{g}_Q\ \hat{g}_{Sw}\ \hat{g}_C\ ]$

(6.35) Tag(L), Conf(L)

Plus the gating record:

(6.36) JumpGate: pass/fail , KLGate: pass/fail , RecoveryCheck: pass/fail

This is the “first usable gain artifact”: compact, falsifiable, and immediately usable for later sections (coupling κ, basis loops, control planning).

---

7. Coupling κ and the basis graph

Up to now we treated each loop **L** as an “operational object” with:

- a loop tube and return map (Section 3),
- a compressed coordinate $\Xi(L)=(\rho,\gamma,\tau)$ (Section 4),
- a local gain artifact $\hat{G}(L)$ and a dominant archetype tag $Tag(L)\in\{G,E,W,S\}$ (Section 6).

This section upgrades from “single loop” to **multi-loop / multi-object** systems by defining a **time-windowed causal coupling** κ and building the **basis graph**.

7.1 Notation: objects, loops, and loop-level coordinates

Let an object A have a set of validated loops:

(7.1) $\mathcal{L}(A) = \{ L_A^1, L_A^2, \ldots, L_A^{m_A} \}$

Each loop has a compiled coordinate and a gain artifact:

(7.2) $\hat{\Xi}(L_A^i) = (\hat{\rho},\hat{\gamma},\hat{\tau})$, $\hat{G}(L_A^i) \in \mathbb{R}^{3\times4}$

and a dominant archetype tag:

(7.3) $Tag(L_A^i) \in \{G,E,W,S\}$

For any time window W, define the window-estimated loop coordinate:

(7.4) $\hat{\Xi}_A^i(W) = \hat{\Xi}(L_A^i; W)$

meaning: “compute Ξ proxies using samples inside window W, restricted to loop tube of L_A^i, passing the same gates as Section 4.”

7.2 κ as time-windowed causal influence: the intervention-first definition

Protocol 2 is “marketing-friendly but hard-nosed”: **κ is defined by controllable perturbations whenever possible.**
The cleanest definition is: *change something in B (via a known operator channel), observe the induced change in A.*

7.2.1 κ for a specific operator channel in B

Fix:

- a loop L_B^j in object B (the “source loop”),
- a loop L_A^i in object A (the “target loop”),
- an operator channel c ∈ {P,Q,Sw,C} applied to B,
- a lag ℓ ≥ 0 and a response window duration Δ > 0.

Run a one-channel intervention on B inside a source window W_src, and measure A in a lagged target window W_tgt:

(7.5) W_tgt = W_src shifted by (ℓ,Δ)

Define the operator step:

(7.6) u_B^c = u_B^c,0 + δu on W_src, and u_B^c = u_B^c,0 otherwise

Then define the **time-windowed causal influence** as a difference-in-differences slope:

(7.7) κ̂_{A←B}^{i←j,c}(ℓ,Δ) = [(Ξ̂_A^i(W_tgt) − Ξ̂_A^i(W_tgt,base))] / δu

where W_tgt,base is a matched “no-intervention” target window (same protocol, same boundary, similar baseline conditions).

Interpretation: κ̂^{i←j,c} is a **3×1 vector** (effect on ρ,γ,τ) per unit operator change in B.

> Minimal matching rule for W_tgt,base (practical): choose a baseline window adjacent in time, or use a library of baseline windows matched by (loop occupancy, leakage rate, recurrence time, and external covariates).

7.2.2 Kernelized (lag-aggregated) κ

Often influence is distributed across lags. Use a declared lag-kernel w(ℓ)≥0 with Σ_ℓ w(ℓ)=1:

(7.8) κ̂_{A←B}^{i←j,c}(Δ) = Σ_{ℓ∈Λ} w(ℓ) κ̂_{A←B}^{i←j,c}(ℓ,Δ)

This converts “many-lag measurements” into a single channel influence.

7.3 Regime gates for κ (don’t confuse coupling with switching)

Coupling estimation is only meaningful if both:

- the target loop in A remains valid during W_tgt, and
- the intervention window in B is not itself dominated by an untracked regime jump (unless that is exactly the mechanism you want to measure).

We reuse gates from Section 6 and add coupling-specific ones.

Gate κ-Go (loop validity on both ends)

Only estimate κ if both loops pass loop validity (Section 3) on their respective windows:

(7.9) Valid(L_A^i; W_tgt) ∧ Valid(L_B^j; W_src)

Gate κ-G1 (jump/KL gate on the target)

Reject κ estimates when A undergoes a regime shift during W_tgt:

(7.10) JumpGate_A(W_tgt)=pass and KLGate_A(W_tgt)=pass

If this fails, either:

- segment W_tgt and compute κ only on pre-switch segment, or
- declare the influence as W-like and record it as a **switch-trigger coupling** (see 7.6.3).

Gate κ-G2 (intervention isolation)

You must be able to attribute the effect to B's channel c. Require:

(7.11) all other operator channels in B are constant (or logged and regressed out) on W_src

---

7.4 Observational fallback: κ without direct interventions (clearly labeled)

Sometimes you cannot intervene on B. Then κ becomes an **estimated directed dependence** in Ξ-space. This is less clean but still useful if you label it as observational.

7.4.1 Linear lagged model (Granger-style, loop-level)

Let Ξ_A^i[n] and Ξ_B^j[n] be block-averaged loop coordinates per block n. Fit:

(7.12) $\Xi_A^i[n+1] = \alpha + \Sigma_{r=0}^{p} A_r \Xi_A^i[n-r] + \Sigma_{r=0}^{p} B_r \Xi_B^j[n-r] + \varepsilon[n]$

Define an observational coupling strength as:

(7.13) $\hat{\kappa}_{A\leftarrow B,obs}^{i\leftarrow j} = \Sigma_{r=0}^{p} \|B_r\|_F$

($\|\cdot\|_F$ is Frobenius norm.)

Rule: always report this as "obs" and do not mix it with intervention κ in the same headline number.

7.5 From loop-to-loop κ to an object coupling map

For each ordered loop pair (i←j) and channel c, $\hat{\kappa}^{i\leftarrow j,c}$ is a 3-vector. For graph building we typically want a scalar edge weight.

7.5.1 Channel-specific scalarization

Define a weighted norm across Ξ components with declared weights w_ρ,w_γ,w_τ:

(7.14) $\|v\|_\Xi = \sqrt{(w_\rho v_\rho^2 + w_\gamma v_\gamma^2 + w_\tau v_\tau^2)}$

Then define the channel edge weight:

(7.15) $w_{A\leftarrow B}^{i\leftarrow j,c} = \| \hat{\kappa}_{A\leftarrow B}^{i\leftarrow j,c}(\Delta) \|_{\Xi}$

7.5.2 Aggregate edge weight across channels

Option A (max channel, for interpretability):

(7.16) $w_{A\leftarrow B}^{i\leftarrow j} = \max_{c\in\{P,Q,Sw,C\}} w_{A\leftarrow B}^{i\leftarrow j,c}$

Option B (sum channel, for "total influence"):

(7.17) $w_{A\leftarrow B}^{i\leftarrow j} = \Sigma_{c} w_{A\leftarrow B}^{i\leftarrow j,c}$

Record also which channel attains the max:

(7.18) $c^*(i\leftarrow j) = \operatorname{argmax}_c w_{A\leftarrow B}^{i\leftarrow j,c}$

This gives you a crisp statement like: "B influences A mainly through Switch-like coupling."

7.6 Build the coupling graph

7.6.1 Loop coupling graph

Define a directed weighted graph $\mathcal{G}$ whose nodes are all validated loops across all objects:

(7.19) $V(\mathcal{G}) = \cup_A \mathcal{L}(A)$

Add a directed edge $L_B^j \rightarrow L_A^i$ if the estimated coupling exceeds a threshold:

(7.20) $\text{edge}(j\rightarrow i) \text{ exists} \Leftrightarrow w_{A\leftarrow B}^{i\leftarrow j} \geq \theta_{\kappa}$

Edge attributes include:

- weight $w_{A\leftarrow B}^{i\leftarrow j}$,
- dominant channel $c^*(i\leftarrow j)$,
- lag window parameters $(\Delta,\Lambda,w(\ell))$,
- confidence (repeatability across trials).

7.6.2 Collapse to an object-level coupling graph (optional)

If you want object-level instead of loop-level, aggregate all edges between objects:

(7.21) $w_{A\leftarrow B} = \Sigma_{i\in\mathcal{L}(A)} \Sigma_{j\in\mathcal{L}(B)} w_{A\leftarrow B}^{i\leftarrow j}$

or use max if you want "any strong coupling":

(7.22) $w_{A\leftarrow B} = \max_{i,j} w_{A\leftarrow B}^{i\leftarrow j}$

7.6.3 Switch-trigger coupling (special case, W-like)

Sometimes B's intervention doesn't "nudge" A; it **triggers a loop transition** in A (a regime

switch). Then κ should be recorded as an effect on hazard, not on mean Ξ.

Define A's loop-transition probability out of L_A^i during W_tgt:

(7.23) p_out^A(i; W_tgt) = ℙ(exit from R_i within W_tgt | start in R_i)

Then define switch-trigger coupling as:

(7.24) κ̂_{A←B,sw}^{i←j,c}(ℓ,Δ) = [p_out^A(i; W_tgt) − p_out^A(i; W_tgt,base)] / δu

This is the correct object when the influence is *discrete switching* rather than continuous deformation.

7.7 Choose basis loops from the coupling graph

We want a **minimal spanning set** of loops that represent the system's persistent operating modes and its coupling skeleton.

7.7.1 Basis by recurrent components (graph-theoretic basis)

Compute the strongly connected components (SCCs) of $\mathcal{G}$. A SCC is **closed** if it has no outgoing edges above θ_κ.

Let the closed SCCs be {$\mathcal{C}_1$,…,$\mathcal{C}$_M}. Define the recurrent basis:

(7.25) $\boldsymbol{\mathcal{B}}$_recur = { pick one representative loop from each $\mathcal{C}$_m }

A principled representative choice is the "most central" loop by weighted in+out strength:

(7.26) centrality(L) = Σ_{incoming} w + Σ_{outgoing} w

Pick argmax centrality within each SCC.

This basis is minimal in the operational sense: each closed SCC is a persistent "mode cluster," and you need at least one loop to represent it.

7.7.2 Basis by signature span (compression basis; optional but powerful)

For each loop L define a signature vector s(L) (from Sections 2–6), e.g.:

(7.27) s(L) = (ρ̂,γ̂,τ̂, D_G,D_E,D_W,D_S, Conf, …)

Choose a minimal set $\boldsymbol{\mathcal{B}}$_sig such that all other loops are approximable:

(7.28) for each L, min_{α_k≥0} ‖ s(L) − Σ_{k∈$\boldsymbol{\mathcal{B}}$_sig} α_k s(L_k) ‖ ≤ ε

In practice: start from $\boldsymbol{\mathcal{B}}$_recur and greedily add loops that reduce the worst approximation error.

7.8 Record dominant force tags per loop (and per edge)

Finally, every node and edge in $\mathcal{G}$ should carry "what kind of mechanism dominates" in a way

compatible with the canonical pairing.

Node tag (loop archetype)

From Section 6, each loop already has:

(7.29) Tag(L) ∈ {G,E,W,S} , Conf(L) ∈ [0,1]

Edge tag (coupling mechanism)

Each edge has a dominant operator channel c*(i←j) from (7.18). Map it to archetype:

(7.30) EdgeTag(j→i) = map(c*(i←j)) , with map(P)=G, map(Q)=E, map(Sw)=W, map(C)=S

This creates an immediately readable "mechanism-labeled network":

- node tags say what stabilizes the loop,
- edge tags say how influence propagates.

7.9 Output artifact: the Basis Graph Card

The output of Section 7 is a portable artifact:

- a loop graph $\mathcal{G}$ with weighted directed edges (7.20),
- a chosen basis loop set $\boldsymbol{\mathcal{B}}$ (7.25, optionally refined by 7.28),
- per-loop node labels (Tag, Conf, $\hat{\Xi}$),
- per-edge labels (weight, dominant channel, lag kernel, gate outcomes),
- a list of "switch-trigger couplings" (7.24) separately tracked.

---

8. Falsifiability harness: turning "maybe similar" into "engineering-testable"

A universal "viewpoint" becomes engineering-grade only if it comes with **hard rejection tests** and **explicit failure routing**. This section defines a minimal harness with four gates:

- **Gate 1:** proxy drift
- **Gate 2:** boundary accounting
- **Gate 3:** probe backreaction
- **Gate 4:** control effectiveness

and a failure router that outputs one of five actions:

(8.1) Route ∈ { shrink boundary, change proxy, split loop, change timescale, elevate model class }

8.0 Harness inputs and log requirements

For each validated loop L (Section 3) and its Ξ proxies (Section 4), you must be able to compute on any window W:

- $\hat{\Xi}(W) = (\hat{\rho}(W), \hat{\gamma}(W), \hat{\tau}(W))$
- loop validity metrics: recurrence, map stability (residual_F, $\hat{L}$_F), leakage $\hat{\ell}$_R(W)
- jump and KL metrics: JumpGate(W), KLGate(W) (Section 6)
- operator logs: u(t) with channel IDs P/Q/Sw/C
- (optional but recommended) at least one "probe signature" observable $\hat{E}(W)$ (e.g., current magnitude M_E or circulation score)

The harness is designed to be **protocol-internal**: it does not assume external ground truth.

8.1 Gate 1: Proxy drift gate (Ξ must be stable under "no real change")

Purpose

Reject cases where Ξ is a moving target because proxies are not stable (measurement artifacts, hidden confounders, regime mixing).

Definition 8.1 (Window drift metrics)

Split an observation period into K windows $\{W_1,\ldots,W_K\}$ where:

- no operator interventions occur (u(t) constant),
- no major exogenous changes are declared.

Compute:

(8.2) $\hat{\mu}_\Xi = (1/K)\,\Sigma_{k=1}^{K}\,\hat{\Xi}(W_k)$

(8.3) $\hat{\Sigma}_\Xi = (1/(K-1))\,\Sigma_{k=1}^{K}\,(\hat{\Xi}(W_k) - \hat{\mu}_\Xi)(\hat{\Xi}(W_k) - \hat{\mu}_\Xi)^T$

Define relative drift (componentwise CV):

(8.4) $CV_\rho = \mathrm{std}(\{\hat{\rho}(W_k)\})/(|\mathrm{mean}(\{\hat{\rho}(W_k)\})|+\varepsilon_0)$
(8.5) $CV_\gamma = \mathrm{std}(\{\hat{\gamma}(W_k)\})/(|\mathrm{mean}(\{\hat{\gamma}(W_k)\})|+\varepsilon_0)$
(8.6) $CV_\tau = \mathrm{std}(\{\hat{\tau}(W_k)\})/(|\mathrm{mean}(\{\hat{\tau}(W_k)\})|+\varepsilon_0)$

Gate condition:

(8.7) $CV_\rho \le c_\rho \wedge CV_\gamma \le c_\gamma \wedge CV_\tau \le c_\tau$

Pass/fail interpretation

- Pass: proxies are stable enough to treat Ξ as an effective coordinate.

- Fail: Ξ is not well-defined under the current protocol.

8.2 Gate 2: Boundary accounting gate (the loop must not “leak reality”)

Purpose

Reject cases where the chosen boundary B fails: too much “outside” influence bleeds in, or the loop is not persistent enough to be an object.

This gate uses leakage and survival, and forces you to admit when your system definition is wrong.

Definition 8.2 (Leakage and survival checks)

Compute leakage rate $\hat{\ell}_R(W)$ in each window (Section 3), and survival over m cycles:

(8.8) $\hat{\ell}_R(W) = 1/(\Delta \cdot \mathbb{E}[\tau_{exit} \mid W])$
(8.9) $\hat{S}_W(m) = \mathbb{P}(\tau_{exit} > mT_{cycle} \mid W)$

Gate condition:

(8.10) $\hat{\ell}_R(W) \le \ell_{max}$ for all W in the evaluation period
(8.11) $\hat{S}_W(m_0) \ge 1 - \varepsilon_{surv}$ for all W

Optional “boundary shock” detector:
if an external disturbance proxy e(t) is logged (temperature, market volatility, etc.), require that leakage does not spike without explanation:

(8.12) $\mathrm{corr}(\hat{\ell}_R(W_k), \bar{e}(W_k)) \le r_{bdry,max}$ (unless declared)

Pass/fail interpretation

- Pass: boundary B yields a persistent object.
- Fail: your “object on the table” is not actually closed enough to support loop-based compression.

8.3 Gate 3: Probe backreaction gate (measurement must not secretly be Pump/Switch/Couple)

Purpose

Probe is an interface/measurement knob (Section 5). If “probing” changes the system as much as Pump/Switch/Couple, then the model must explicitly account for observer backreaction (or you must redefine the protocol).

We test whether small Probe changes cause **unintended** large changes in (ρ,γ,τ) or induce regime jumps.

Definition 8.3 (Backreaction metrics)

Run a “null probe”: a small Probe perturbation δu_Q that is intended to be minimally invasive (declared). Compute:

(8.13) $\Delta\hat{\Xi}_Q = \hat{\Xi}_{after} - \hat{\Xi}_{before}$ under null probe

Also compute jump and KL changes:

(8.14) $\Delta\hat{r}_J = \hat{r}_J(after) - \hat{r}_J(before)$
(8.15) $D_{sym} = 1/2\ KL(\hat{P}_{after} \| \hat{P}_{before}) + 1/2\ KL(\hat{P}_{before} \| \hat{P}_{after})$

Gate conditions:

(8.16) $\|\Delta\hat{\Xi}_Q\| \le \varepsilon_{probe}$
(8.17) $\Delta\hat{r}_J \le r_{J,probe,max}$
(8.18) $D_{sym} \le \kappa_{KL,probe}$

Pass/fail interpretation

- Pass: Probe can be treated as a measurement/interface channel with limited backreaction.
- Fail: probing is not “just observation”; it is dynamically invasive and must be modeled as such.

8.4 Gate 4: Control effectiveness gate (operators must predictably move Ξ)

Purpose

A framework that cannot produce repeatable operator→response is not engineering-testable. This gate checks whether MEP gains are real and reusable.

Definition 8.4 (Repeatability and sign-consistency)

From Section 6 you estimated a gain matrix $\hat{G}$ from one-channel trials. Repeat the same MEP trial R times (same loop, same protocol, same δu_i) across separated times.

Let $\hat{g}_i^{(r)}$ be the estimated gain column for channel i in repetition r. Define mean and dispersion:

(8.19) $\bar{g}_i = (1/R) \Sigma_{r=1}^{R} \hat{g}_i^{(r)}$
(8.20) $Var(g_i) = (1/(R-1)) \Sigma_{r=1}^{R} \|\hat{g}_i^{(r)} - \bar{g}_i\|^2$

Gate conditions (minimal):

(i) Effect size: at least one channel should have nontrivial effect:

(8.21) $\max_i \|\bar{g}_i\| \ge g_{min}$

(ii) Repeatability: effect is not pure noise:

(8.22) $Var(g_i) \le v_{max} \|\bar{g}_i\|^2$ for channels with $\|\bar{g}_i\| \ge g_{min}$

(iii) Sign-consistency for paired coordinate (recommended):

(8.23) $sign((\hat{g}_P)_\rho)$ stable across repetitions

(8.24) sign($(\hat{g}_C)_\gamma$) stable across repetitions
(8.25) sign($(\hat{g}_{Sw})_\tau$) stable across repetitions

Pass/fail interpretation

- Pass: you have a reusable "gain artifact" to do prediction and control.
- Fail: either the system is too nonstationary, your loop is mixed, or your timescale is wrong.

8.5 Failure routing rules (what to do when a gate fails)

A falsifiability harness must not only say "fail"; it must output an action. The router uses which gate failed plus diagnostic indicators to choose a route in (8.1).

8.5.1 Router inputs

Let the boolean gate results be:

(8.26) G1 = ProxyDriftPass
(8.27) G2 = BoundaryAccountingPass
(8.28) G3 = ProbeBackreactionPass
(8.29) G4 = ControlEffectivenessPass

Also keep supporting diagnostics:

- leakage $\hat{\ell}_R$, survival $\hat{S}$
- KL/jump metrics
- return-map drift ε_{drift}, Lipschitz $\hat{L}_F$
- sample adequacy (cycles and events count)

8.5.2 Routing logic (explicit, deterministic defaults)

Case A: Gate 2 fails (boundary problem dominates)
Typical symptoms: high leakage, low survival, loop breaks frequently.

Route:

(8.30) if not G2 ⇒ Route = shrink boundary

Interpretation: your "object" is too porous; reduce scope or redefine what is inside B.

Case B: Gate 1 fails but Gate 2 passes (proxy problem dominates)
Symptoms: object persists, but Ξ proxies drift under no intervention.

Route:

(8.31) if (not G1) and G2 ⇒ Route = change proxy

Common fixes:

- replace density-based $\hat{\rho}_\Phi$ with exit-time proxy $\hat{\rho}_{exit}$,

- replace constraint-based $\hat{\gamma}_{conf}$ with survival-only $\hat{\gamma}_{simple}$,
- replace $\hat{\tau}_{rec}$ from return map with $\hat{\tau}$ from recurrence time.

Case C: Gate 3 fails (observer coupling is dynamically invasive)
Symptoms: "null probe" triggers KL shift or jump spikes, or large $\Delta\Xi$.

Route:

(8.32) if not G3 ⇒ Route = elevate model class

Meaning: you must include explicit observer/interface dynamics (e.g., augment state with y(t), non-commuting measurements, or a GKSL/Lindblad-like effective description). If you do not want to elevate, then redefine Probe channel and treat it as Pump/Switch/Couple (i.e., admit it is an intervention, not observation).

Case D: Gate 4 fails with strong jump/KL evidence (loop mixing / wrong segmentation)
Symptoms: repeated interventions yield inconsistent gains because the system keeps switching modes; KL and jump gates trigger often.

Route:

(8.33) if (not G4) and (JumpGate frequently fails or KLGate frequently fails) ⇒ Route = split loop

Meaning: what you called one loop is actually multiple loops/regimes. Rebuild loop tubes R_i and repeat compilation.

Case E: Gate 4 fails without strong jump/KL evidence (timescale mismatch)
Symptoms: loop seems stable, no big KL shifts, but gains are noisy and non-repeatable. Often because windows are too short/long relative to the true response time.

Route:

(8.34) if (not G4) and (JumpGate mostly passes) and (KLGate mostly passes) ⇒ Route = change timescale

Meaning: adjust Δ, window lengths $T_0/T_1/T_2$, and the lag kernel in κ estimation.

8.5.3 Tie-break rule (when multiple gates fail)

If multiple gates fail, apply priority:

(8.35) Boundary (G2) > Probe backreaction (G3) > Proxy drift (G1) > Control effectiveness (G4)

Reason: if the object/boundary is wrong or the probe is invasive, fixing proxies or gains first is wasted effort.

8.6 Harness artifact: the Harness Checklist Card

For each loop L, record:

(8.36) Gate1 ProxyDrift: pass/fail with (CV_ρ,CV_γ,CV_τ)
(8.37) Gate2 Boundary: pass/fail with (ℓ̂_R, Ŝ(m0))
(8.38) Gate3 ProbeBackreaction: pass/fail with (‖ΔΞ̂_Q‖, D_sym, Δr̂_J)
(8.39) Gate4 ControlEffectiveness: pass/fail with (g_min, Var ratios, sign stability)
(8.40) FailureRoute(L) ∈ {shrink boundary, change proxy, split loop, change timescale, elevate model class}

This is what turns “maybe similar” into a **testable engineering protocol**: failures do not create narrative; they create **actions**.

---

9. Standard artifacts (copy-paste templates)

This section defines the **canonical reporting artifacts** of Protocol 2. They are designed to be:

- **portable** across domains (physics, biology, orgs, finance),
- **reproducible** (protocol-bound, with gates and parameter declarations),
- **safe to publish** (no “this replaces QM” misunderstanding).

Artifacts:

1. **Loop Card** (per loop)
2. **Coupling Card** (per directed loop-pair, optionally per channel)
3. Minimal outside-facing explanation script

Everything below is intended as copy-paste templates.

9.1 Loop Card (canonical)

LOOP CARD — Template

[Loop ID] L = (Object=A, LoopName=..., Version=...)

Protocol P = (B, Δ, h, u)

- Boundary B: ...
- Sampling step Δ: ...
- Observation map h: $\mathcal{M}\to\mathcal{Z}$ (summary): ...
- Admissible operator channels u: {Pump, Probe, Switch, Couple}
- Data window(s): start=..., end=..., timezone=...

Loop tube and section

- Tube R ⊆ $\mathcal{Z}$ (definition): ...
- Section Σ (if used): ...

- Nominal cycle period: T^_cycle = …
- Recurrence time: T^_rec = … (Eq. 4.6)

Loop validity (Section 3)

- Recurrence: pass/fail
  - T^_rec = … , threshold T_rec = …
- Return-map stability: pass/fail
  - residual_F = … (Eq. 4.8)
  - L^_F = … (Eq. 4.9)
  - drift bound ε_drift = …
- Bounded leakage: pass/fail
  - ℓ^_R = … (Eq. 4.4)
  - survival Ŝ(m0)=… (Eq. 3.14–3.15)

Generator archetype signatures (Section 2)

- D_G = … , D_E = … , D_W = … , D_S = …
- Supporting signatures (optional):
  - Cir^_E(Γ)=… (Eq. 2.31)
  - r^_W=… (Eq. 4.10)
  - R^_S, N^_S, d_eff(0.95)=… (Eq. 4.19–4.21, 2.38)

Compiled coordinates Ξ^(L) (Section 4)

- ρ^ = … (proxy choice: Φ-depth / exit-time / stock)
- γ^ = … (proxy choice: survival / confinement / +coupling)
- τ^ = … (proxy choice: recovery / switching / max)
- Proxy gates passed: Gate 0=…, ρ-G1=…, γ-G1=…, τ-G1=…, Ξ-G1=…

MEP gain artifact (Section 6)

- Perturbation amplitudes: δu_P=…, δu_Q=…, δu_Sw=…, δu_C=…
- Jump/KL thresholds: θ=…, κ_KL=…
- Gain matrix:

(9.1) Ĝ(L) = [ĝ_P ĝ_Q ĝ_Sw ĝ_C] ∈ ℝ^{3×4}

- Column summaries (with uncertainty if available):
  - ĝ_P = (∂ρ/∂u_P, ∂γ/∂u_P, ∂τ/∂u_P)=…

- ĝ_Q = …
- ĝ_Sw = …
- ĝ_C = …

Loop tag

- Structure tag: Tag_struct = argmax(D_G,D_E,D_W,D_S) = …
- Gain tag: Tag_gain = map(argmax(Ŝ_P,Ŝ_Q,Ŝ_Sw,Ŝ_C)) = …
- Final tag: Tag(L) = … , Conf(L)=… (Eq. 6.30–6.32)

Falsifiability harness (Section 8)

- Gate1 ProxyDrift: pass/fail (CV_ρ=…, CV_γ=…, CV_τ=…)
- Gate2 Boundary: pass/fail (ℓ̂_R=…, Ŝ(m0)=…)
- Gate3 ProbeBackreaction: pass/fail (‖ΔΞ̂_Q‖=…, D_sym=…, Δr̂_J=…)
- Gate4 ControlEffectiveness: pass/fail (repeatability stats=…)
- FailureRoute(L) = … (Eq. 8.30–8.35)

Notes / known limitations

- …

9.2 Coupling Card (canonical)

Coupling is directional and time-windowed. This card can be written at:

- **loop-to-loop level** (recommended default), or
- object-to-object level (aggregation).

COUPLING CARD — Template (loop-to-loop)

[Coupling ID] L_B^j → L_A^i (Source=B:j, Target=A:i)

Protocol and windows

- Protocol P shared? yes/no (if no, specify P_A and P_B separately)
- Source window W_src: start=…, end=…
- Target window W_tgt: start=…, end=…
- Lag model: lags Λ={…}, kernel w(ℓ)=… (Eq. 7.8)
- Baseline matching window W_tgt,base: definition=…

Intervention definition (preferred)

- Operator channel applied on B: c ∈ {P,Q,Sw,C}

- Step amplitude: δu = …
- Operator schedule: u_B^c(t)=… (Eq. 7.6)

κ estimate (vector)

- Windowed influence:

(9.2) κ̂_{A←B}^{i←j,c}(ℓ,Δ) = [Ξ̂_A^i(W_tgt) − Ξ̂_A^i(W_tgt,base)] / δu

- Lag-aggregated:

(9.3) κ̂_{A←B}^{i←j,c}(Δ) = Σ_{ℓ∈Λ} w(ℓ) κ̂_{A←B}^{i←j,c}(ℓ,Δ)

Scalar edge weight (for graphs)

- Ξ-weighted norm:

(9.4) w_{A←B}^{i←j,c} = ‖κ̂_{A←B}^{i←j,c}(Δ)‖_Ξ (Eq. 7.14–7.15)

- Dominant channel (if multi-channel tested):

(9.5) c* = argmax_c w_{A←B}^{i←j,c}

- Edge tag:

(9.6) EdgeTag = map(c*) ∈ {G,E,W,S}

Switch-trigger coupling (special case)
If the effect is primarily to trigger a loop transition in A, record:

(9.7) κ̂_{A←B,sw}^{i←j,c} = [p_out^A(i;W_tgt) − p_out^A(i;W_tgt,base)] / δu

Coupling gates

- κ-G0 loop validity: pass/fail (Eq. 7.9)
- κ-G1 target jump/KL: pass/fail (Eq. 7.10)
- κ-G2 intervention isolation: pass/fail (Eq. 7.11)
- If failed: failure routing (Section 8) suggestion=…

Notes

- Confounders checked: …
- Repeatability: R=… trials; variance=…

9.3 Minimal outside-facing explanation script (safe, avoids “replace QM” alarms)

Below is a compact script you can paste at the top of posts, talks, or wiki pages. It’s intentionally conservative.

Outside-facing script — Template

What this is:
We present an operational protocol for analyzing *open systems with recurrent loops* using a generator-level language (drift–diffusion–jump). The method decomposes observed dynamics into four mechanism archetypes (potential/gradient, circulation/current, switching/jumps, confinement/constraints) and compresses each validated loop into three effective coordinates $\Xi=(\rho,\gamma,\tau)$. We then define four controlled perturbation channels—Pump, Probe, Switch, Couple—and a minimal experiment protocol to estimate local gains and directed coupling between loops, with explicit falsifiability gates and failure-routing rules.

What this is not:
This is not a claim to replace quantum mechanics or fundamental physics. The protocol works at the level of *effective dynamics under a declared boundary and measurement protocol*, similar in spirit to effective theories in many sciences. If the boundary or protocol changes, the loop decomposition and Ξ coordinates may change, and the falsifiability harness is designed to detect that.

Why physicists may still care:
All definitions are generator-based and testable: loops require recurrence, stable return maps, and bounded leakage; gains require repeatable operator-response; and regime changes are rejected using jump/KL gates. The result is a portable, engineering-grade "control viewpoint" for complex systems—usable whether the system is a mechanical oscillator, a biological regulatory loop, or an organizational process.

---

10. Optional Appendix: when you *really* need non-commuting / order effects

This appendix adds a **Layer B** extension you only invoke when **Layer A (classical Markov generator on $\mathcal{M}$)** becomes insufficient—specifically when **Probe operations are non-exchangeable** (order effects), or when the **Probe backreaction gate** (Section 8, Gate 3) fails in a way you can't resolve by redefining proxies/boundaries.

The goal is *engineering adequacy*, not metaphysics: we adopt a **GKSL/Lindblad-like effective generator** purely as a mathematically disciplined way to model **order-dependent interventions**.

10.1 Trigger condition: when Layer A is no longer adequate

Layer B is activated if **any** of the following is observed under a fixed protocol P:

Trigger T1: explicit order effects (non-exchangeable probes)

Choose two probe settings q_1 and q_2 (these are two Probe knob settings, or two distinct probe instruments). Apply them in opposite orders on matched initial conditions and compare outcome distributions.

Let Y be the probe readout (can be discrete labels or a binned continuous output). Define:

(10.1) $P_{12}(y) = \mathbb{P}(Y=y \mid \text{apply } q_1 \text{ then } q_2)$
(10.2) $P_{21}(y) = \mathbb{P}(Y=y \mid \text{apply } q_2 \text{ then } q_1)$

Define the order-effect divergence:

(10.3) D_order = 1/2 KL(P_{12}‖P_{21}) + 1/2 KL(P_{21}‖P_{12})

Order-effect gate:

(10.4) D_order ≥ κ_order ⇒ Layer B required

If D_order is below threshold, Layer A remains acceptable (and you treat Probe as "approximately exchangeable").

Trigger T2: Probe backreaction gate fails irreducibly

If the "null probe" (Section 8, Gate 3) causes large shifts:

(10.5) ‖ΔΞ̂_Q‖ > ε_probe or D_sym > κ_KL,probe or Δr̂_J > r_J,probe,max

and you **cannot** fix it by:

- shrinking boundary,
- changing proxy,
- splitting loop,
- changing timescale,

then you elevate to Layer B. (This is exactly the failure routing rule "elevate model class.")

10.2 Layer B object: state is no longer a point; it is an "information state" ϱ

Layer A assumes the system state can be represented as a point x(t) on $\mathcal{M}$ with a Markov generator $\mathscr{L}$.

Layer B replaces "point state" with an **information state** ϱ(t) that supports non-commuting operations. Minimal choice:

- ϱ is a positive, trace-one operator (a density-operator-like object) on a chosen state space $\boldsymbol{\mathcal{H}}$.

(10.6) ϱ ≥ 0 , Tr(ϱ)=1

Interpretation (outside-facing safe phrasing): ϱ is an *effective state of knowledge / interaction record* needed to model order effects, not a claim about microscopic quantum ontology.

10.3 Layer B dynamics: GKSL/Lindblad-like generator

A standard, minimal, stable class for continuous-time dynamics of ϱ is the GKSL form (after Vittorio Gorini, Andrzej Kossakowski, E. C. G. Sudarshan, Goran Lindblad):

(10.7) dϱ/dt = $\boldsymbol{\mathcal{L}}$_B(ϱ) = −i[H,ϱ] + Σ_k (L_k ϱ L_k† − 1/2 {L_k†L_k, ϱ})

where:

- H is an effective Hamiltonian-like generator (order-sensitive “rotation” component),
- {L_k} are dissipators capturing irreversible coupling, decoherence, coarse-graining, and measurement disturbance,
- [A,B]=AB−BA and {A,B}=AB+BA.

This is the minimal “well-posed” layer that:

- preserves positivity and trace,
- naturally represents **non-commuting** actions,
- reduces to classical stochastic behavior under commuting/diagonal limits.

10.4 How the four operators appear in Layer B (still as generator perturbations)

We keep the exact same control philosophy:

(10.8) $\mathcal{L}_{B,u} = \mathcal{L}_{B,0} + u_P\, \mathcal{L}_{B,P} + u_Q\, \mathcal{L}_{B,Q} + u_{Sw}\, \mathcal{L}_{B,Sw} + u_C\, \mathcal{L}_{B,C} +$ higher-order terms

A practical, conservative mapping consistent with the “marketing hook” is:

Pump (Gravity-like ↔ Pump): reshape potential depth via H

Pump changes the “energy landscape” term:

(10.9) $H(u_P) = H_0 + u_P\, H_P$

so Pump enters as:

(10.10) $\mathcal{L}_{B,P}(\varrho) = -i[H_P, \varrho]$

This is the cleanest Layer B analog of “G-like potential deformation.”

Probe (EM-like ↔ Probe): non-exchangeable instruments as CPTP maps and/or dissipators

In Layer B, **Probe is an instrument** (a family of completely positive maps). For a probe setting q with outcomes y:

(10.11) $\varrho \mapsto \mathcal{M}_{q,y}(\varrho) = M_{q,y}\, \varrho\, M_{q,y}^{\dagger}$

Outcome probability:

(10.12) $\mathbb{P}(y|q) = \mathrm{Tr}(\, M_{q,y}^{\dagger}\, M_{q,y}\, \varrho\,)$

Selective update:

(10.13) $\varrho' = \mathcal{M}_{q,y}(\varrho) \,/\, \mathrm{Tr}(\mathcal{M}_{q,y}(\varrho))$

Unselective (outcome-averaged) probe channel:

(10.14) $\mathcal{E}_q(\varrho) = \Sigma_y \mathcal{M}_{q,y}(\varrho)$

Non-commuting / order effects appear as:

(10.15) $\mathcal{E}_{q1}\circ\mathcal{E}_{q2} \neq \mathcal{E}_{q2}\circ\mathcal{E}_{q1}$

In continuous time, "probing continuously" is commonly modeled as additional dissipators {L_k(u_Q)} that depend on Probe intensity u_Q.

Switch (Weak-like ↔ Switch): jump between generator blocks (piecewise GKSL)

Switch in Layer B is most naturally a **jump process between regimes** $r \in \{1,\ldots,R\}$, each with its own GKSL generator $\mathcal{L}^{(r)}$.

Let r(t) be a Markov jump process with hazard $\lambda_{r\to r'}(u_{Sw})$. Then:

(10.16) $d\varrho/dt = \mathcal{L}^{(r(t))}(\varrho)$

and r(t) jumps with controlled rates:

(10.17) $\lambda_{r\to r'}(u_{Sw}) = \lambda_{r\to r',0} \exp(u_{Sw}\, s_{r\to r'})$

This is the exact Layer B counterpart of "W-like switching."

Couple (Strong-like ↔ Couple): enforce constraints via strong dissipators or penalties

Couple increases closure/binding by adding dissipators that rapidly suppress constraint violations.

Let $\mathbb{N}$ be the "allowed subspace" (or constraint) represented by a projector Π ($\Pi^2=\Pi$). A simple confinement dissipator is:

(10.18) $L_C = \sqrt{u_C}\,(I - \Pi)$

Adding L_C into (10.7) penalizes components outside $\mathbb{N}$. In effect, larger u_C yields stronger confinement and reduces leakage—precisely the S-like role.

10.5 Minimal non-commutation metric you can actually estimate

The diamond norm of channel commutators is the gold standard, but for this paper we want an **engineering metric from logs**.

10.5.1 Observable-level order-effect score

Let f be any scalar statistic you compute from probe outcomes (mean, variance, classification rate, etc.). Define:

(10.19) $\Delta_{order}(f) = |\ \mathbb{E}[f \mid q_1\to q_2] - \mathbb{E}[f \mid q_2\to q_1]\ |$

Aggregate over a small set of declared probe statistics $\{f_1,\ldots,f_m\}$:

(10.20) $S_{order} = \Sigma_{a=1}^{m} w_a \Delta_{order}(f_a)$

Gate:

(10.21) $S_{order} \geq s_{min} \Rightarrow$ Layer B required

This is a practical companion to KL-based D_{order} in (10.3).

10.6 How Layer B connects back to Protocol 2 artifacts (so you don't lose portability)

Layer B **does not replace** the Loop Card / Coupling Card / Harness Card. It only changes what you treat as the "state" and how Probe is modeled.

10.6.1 Loop definition still works

Loops remain protocol-defined recurrences in observed space $\mathcal{Z}$:

- you still validate recurrence, return-map stability, bounded leakage (Section 3),
- but the underlying mechanism generating z is now "instrumented dynamics of ϱ."

10.6.2 Ξ compilation still works (ρ,γ,τ remain loop-level)

You still compile Ξ from loop observables; you just add *optional* Layer-B diagnostics when useful:

- Purity (a proxy for "how collapsed/coherent the state is"):

(10.22) $\mathrm{Pur}(\varrho) = \mathrm{Tr}(\varrho^2)$

- A simple "probe-induced disturbance" score:

(10.23) $\mathrm{Dist}_q = \|\mathcal{E}_q(\varrho) - \varrho\|$ (use a declared matrix norm or proxy observable)

These remain optional and are only invoked if Gate 3 / order tests demand.

10.7 Layer B addendum: the minimal reporting block (copy-paste)

When Layer B is enabled, append this to the Loop Card:

Layer B enabled: yes
Trigger: T1 (order effects) / T2 (probe backreaction)
Order test: D_{order}=… (Eq. 10.3), threshold κ_{order}=…
Probe channel non-exchangeability: q_1,q_2 listed, statistics used listed
Layer B model class: GKSL piecewise / continuous

- H terms used: H_0, H_P (Eqs. 10.9–10.10)
- Dissipators used: $\{L_k\}$ listed (Eq. 10.7)

- If switching: regimes r=..., hazards λ_{r→r′} (Eq. 10.17)
  Interpretation note (outside-facing): "Effective non-commuting probe model; not a claim about microphysical QM replacement."

Minimal outside-facing line (safe)

If you need one sentence:

Layer B is an optional effective modeling upgrade used only when probe operations are empirically non-exchangeable; it adopts a GKSL/Lindblad-like generator to represent order-dependent backreaction without making any claim about replacing fundamental quantum theory.

Conclusion

Protocol 2 gives a **bottom-up, physics-legible** path to a universal operational viewpoint without making any ontological claims.

You start from a minimal dynamical object that physicists accept: an **open stochastic system** with a well-defined generator $\mathscr{L}$ (drift + diffusion + jumps) under a declared boundary and sampling protocol. You then require **loop existence**—recurrence, return-map stability, and bounded leakage—so "objecthood" is not a metaphor but a **testable closure property**.

From there, the method proceeds in three compress–control–verify steps:

1. **Compress:** for each validated loop, compile a protocol-dependent but measurable triple
   (C.1) $\Xi(L) = (\rho,\gamma,\tau)$
   where ρ is a basin/stock depth proxy (G-dominant sensitivity), γ is a closure/exit-cost proxy (S-dominant sensitivity), and τ is a recovery/switching timescale proxy (W-dominant sensitivity), each guarded by identifiability and stability gates.

2. **Control:** formalize four operators as **controlled perturbations of the generator**—Pump, Probe, Switch, Couple—and use a Minimal Experiment Protocol (MEP) to estimate a first usable **local gain artifact**
   (C.2) $\Delta\Xi \approx G\,\Delta u$
   with jump/KL regime rejection so gains are not falsely fitted across mode changes. The canonical pairing remains precise at generator level:
   (C.3) Gravity-like ↔ Pump, EM-like ↔ Probe, Weak-like ↔ Switch, Strong-like ↔ Couple.

3. **Verify and scale:** define directed coupling $\kappa_{A\leftarrow B}$ as a time-windowed causal influence estimate between loops/objects, build a coupling graph, choose **basis loops** as a minimal spanning set, and label nodes/edges by dominant archetype tags. Throughout, a falsifiability harness converts "maybe similar" into "engineering-testable" via four hard gates (proxy drift, boundary accounting, probe backreaction, control effectiveness) plus explicit failure routing rules (shrink boundary, change proxy, split loop, change timescale, elevate model class).

The result is a portable reporting system built around two canonical artifacts:

- **Loop Card:** loop validity + Ξ proxies + gain matrix + tags + harness outcomes.
- **Coupling Card:** κ estimates + gates + dominant coupling mechanism tags.

These artifacts let you compare systems across domains using the same operational language—**not because all systems are the same**, but because the protocol only claims that **loop-**

bearing open systems admit an effective control description under a declared boundary and measurement regime.

What is claimed (falsifiable scope)

- A **generator-level decomposition** into four mechanism archetypes (potential/gradient, circulation/current, switching/jumps, confinement/constraints) that is computable from trajectories and signatures.
- A **loop-based object criterion** that can be accepted or rejected from data.
- A **3-coordinate compression** $\Xi=(\rho,\gamma,\tau)$ compiled from loop observables with explicit stability gates.
- A **minimal experiment routine** that outputs repeatable local gain artifacts and directed coupling estimates (or fails with actionable routing).
- A **standard reporting format** (Loop Card / Coupling Card) enabling cross-domain comparison without narrative drift.

What is not claimed (explicitly)

- No claim that this is a new microphysical theory or that it replaces quantum mechanics.
- No claim that the decomposition is unique independent of protocol; Ξ is **boundary- and observer-protocol dependent** by design.
- No claim of universal completeness; if the harness fails, the method instructs you to refine boundary/proxies/timescale or elevate the model class.

The optional extension

When probes are empirically **non-exchangeable** (order effects) or when probe backreaction cannot be made small, Layer B provides a disciplined upgrade: a GKSL/Lindblad-like effective generator for order-dependent dynamics. This is invoked only by explicit order tests and is reported as an effective modeling choice, not an ontological statement.

Reference

This article intended to use a different approach to present similar contents of the following article, which show the original spark / invention pathway for this particular framework.

- The Ξ-Stack Universal Viewpoint: A Two-Layer (Σ/Ξ) Operator Playbook in (ρ, γ, τ) with Probe–Couple–Pump–Switch Control and a Falsifiability Harness https://osf.io/nq9h4/files/osfstorage/699a231140e75f3888505f47

© 2026 Danny Yeung. All rights reserved. 版权所有 不得转载

Disclaimer

This book is the product of a collaboration between the author and OpenAI's GPT-5.2, X's Grok language model. While every effort has been made to ensure accuracy, clarity, and insight, the content is generated with the assistance of artificial intelligence and may contain factual,

interpretive, or mathematical errors. Readers are encouraged to approach the ideas with critical thinking and to consult primary scientific literature where appropriate.

This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.

I am merely a midwife of knowledge.